



SYDNEY BOYS HIGH SCHOOL

2026

YEAR 12
TASK 4
TRIAL HSC

Name: _____

NESA Number: _____

--	--	--	--	--	--	--	--	--

Tick next to your teacher's name

AF	☐	PSP	☐
JJ	☐	AYW	☐
RCW	☐	JT	☐
		MZ	☐

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Only NESA approved calculators may be used
- A reference sheet is provided with this paper
- All answers, unless otherwise stated, should be left in simplified, exact form
- Marks may **NOT** be awarded for messy or badly arranged work
- For questions in Section II, show ALL relevant mathematical reasoning and/or calculations as answers that rely totally on calculator technology may not necessarily receive full marks.

Total Marks: 70

Section I – 10 marks (pages 2 – 8)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 9 – 16)

- Attempt all Questions in Section II
- Allow about 1 hour and 45 minutes for this section

Examiner:

External Examiner

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

Question 1

Which of the following is equal to $\int \frac{1}{\sqrt{4-9x^2}} dx$?

A. $\frac{1}{3} \sin^{-1} \frac{3x}{2} + C$

B. $\frac{1}{6} \sin^{-1} \frac{3x}{2} + C$

C. $\frac{1}{3} \sin^{-1} \frac{x}{2} + C$

D. $\frac{1}{6} \sin^{-1} \frac{x}{2} + C$

Question 2

A 100 litre tub of salt water has initial concentration 3 g/L.

Salt water of concentration 2 g/L flows in at 1 L/min and the mixture is stirred continuously, before flowing out of the tub at 1 L/min. Which of the following correctly gives the equation for the rate of change of concentration c in g/L at time t minutes?

A. $\frac{dc}{dt} = 2 - \frac{c}{100}$

B. $\frac{dc}{dt} = \frac{1}{100} - 2c$

C. $\frac{dc}{dt} = \frac{1}{50} - c$

D. $\frac{dc}{dt} = \frac{1}{50} - \frac{c}{100}$

Question 3

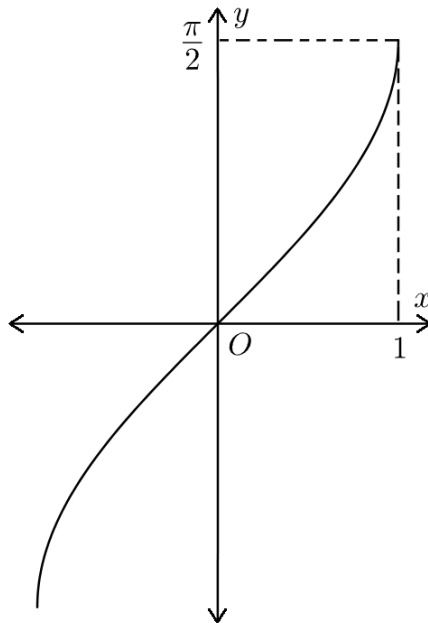
Let $P(x)$ be a polynomial that leaves a remainder of 2 when divided by $x - 3$ and a remainder of -2 when divided by $x + 1$.

Which of the following is the remainder of $P(x)$ when it is divided by $x^2 - 2x - 3$?

- A. $-x + 1$
- B. $-x - 1$
- C. $x + 1$
- D. $x - 1$

Question 4

The graph shows a sketch of $y = \sin^{-1} x$.



The graph is shifted down by $\frac{\pi}{2}$ units, before rotating around the region bounded by the curve, the y -axis and $y = 0$, around the x -axis to form a volume of revolution.

Which of the following is equal to the volume of the solid?

- A. $\pi \int_0^1 \left(\frac{\pi}{2} - \sin^{-1} x \right)^2 dx$
- B. $\pi \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \sin x \right)^2 dx$
- C. $\pi \int_0^1 \left[\frac{\pi}{2} - (\sin^{-1} x)^2 \right] dx$
- D. $\pi \int_0^{\frac{\pi}{2}} \left[\frac{\pi}{2} - (\sin x)^2 \right] dx$

Question 5

A test contains 5 questions, each question can score 1 point if it is answered correctly, 0 if not answered, and will deduct a point from the total score if it is answered incorrectly.

The probability of a student answering a random question correctly is 0.3.

The student will also leave on average, one out of ten questions blank.

A score of at least 3 is required to pass.

What is the probability that the student will pass?

- A. 0.00837
- B. 0.03348
- C. 0.06615
- D. 0.1638

Question 6

Which of the following is equivalent to the function $f(\theta) = \cos^{-1}(\cos \theta)$ defined over the domain $\theta \in [0, 2\pi]$?

- A. $\pi + |\pi - \theta|$
- B. $\pi + |\pi + \theta|$
- C. $\pi - |\pi - \theta|$
- D. $\pi - |\pi + \theta|$

Question 7

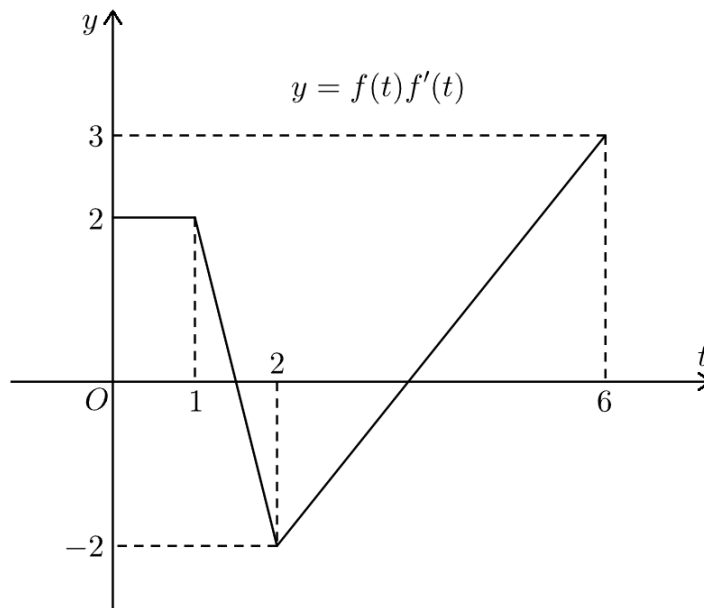
9 position vectors all have length 6 units. At least two of the position vectors will have a minimum dot product equal to which of the following?

- A. 0
- B. $18\sqrt{2}$
- C. $18\sqrt{3}$
- D. 36

Question 8

A particle's velocity is a function of time $v = f(t)$.

The diagram shows the graph of $y = f(t)f'(t)$.



The particle has an initial velocity of 3 m/s.

Calculate the speed of the particle at time $t = 6$.

- A. 3 m/s
- B. 4 m/s
- C. $\sqrt{13}$ m/s
- D. $\sqrt{17}$ m/s

Question 9

Let $f(x) = (2x + 1)(3x + 1)(4x + 1) \dots (20x + 1)$.

Which of the following is equal to $f'(0)$?

A. 207

B. 208

C. 209

D. 210

Question 10

Let $y = f(x)$ be a continuous function defined for all real x .

It has an inverse function defined by $y = g(x)$.

Which of the following is equal to the derivative of $y = g(g(x))$?

A. $\frac{1}{f'(g(x))f'(g^2(x))}$

B. $\frac{1}{f'(x)f'(g(x))}$

C. $\frac{1}{g'(f(x))g'(f^2(x))}$

D. $\frac{1}{g'(x)g'(f(x))}$

Section II

Attempt Questions 11 – 14

Allow 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet.

Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (14 marks) Use a SEPARATE writing booklet.

(a) Solve $\frac{2x+1}{3-x} \geq x$. **2**

(b) Let α, β, γ be the roots of the polynomial $3x^3 - 3x^2 + 4x + 9 = 0$. **2**
Evaluate $\alpha^2 + \beta^2 + \gamma^2$.

(c) Evaluate the integral $\int_0^{\frac{\pi}{2}} \sin \theta \sin^2 2\theta \, d\theta$. **3**

(d) Find the coefficient of x in the expansion of $(1 - 3x)(2x - 1)^9$. **2**

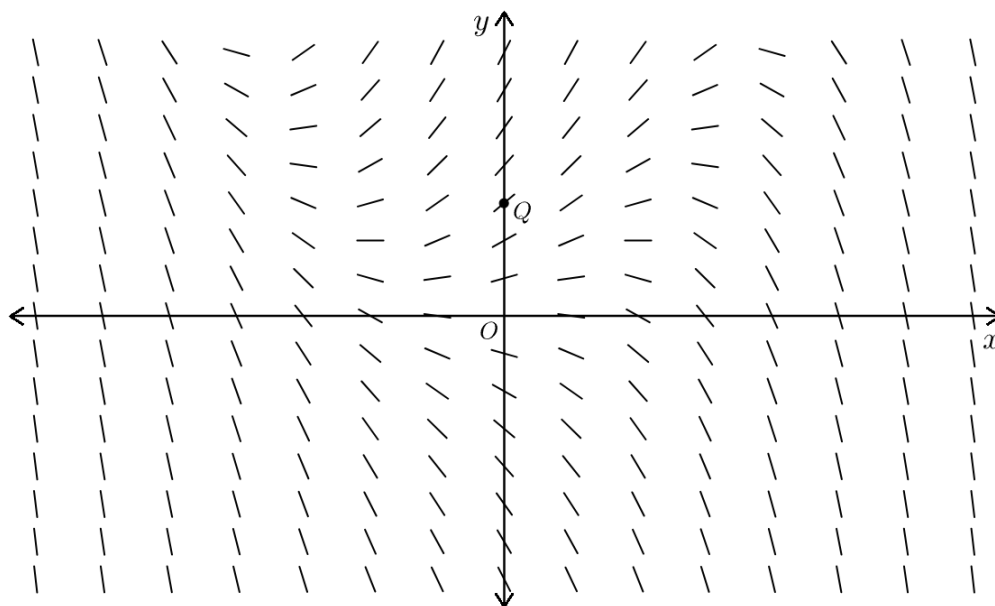
(e) Prove that $\tan\left(\frac{\pi}{4} + \theta\right) = \sec 2\theta + \tan 2\theta$ by using the substitution $t = \tan \theta$. **3**

(f) It is given that a and b are real numbers such that $\sin^{-1} a = \cos^{-1} b$. **2**
Prove that (a, b) lie on the unit circle in the first quadrant.

Question 12 (15 marks)

Use a SEPARATE writing booklet.

- (a) A section of the differential equation $\frac{dy}{dx} = y - x^2$ is shown, along with a point Q .

2

On the sheet provided, sketch a possible solution curve that passes through the point Q .

- (b) Let n be an odd positive integer.

3

Prove by mathematical induction that $5^n + 2^n$ is divisible by 7.

- (c) Some butter is in the shape of a sphere, with radius r .

It is placed into an oven so that its volume is decreasing at a rate proportional to the square root of its surface area, S . You may assume as the butter melts, it is melting uniformly along its surface.

- (i) Show that $\frac{dr}{dt} = \frac{-k}{\sqrt{S}}$ for some positive constant $k > 0$.

1

- (ii) The butter initially had radius 2 cm. By how much does the total melting time exceed the time needed for the butter to shrink to half of its original radius?

3

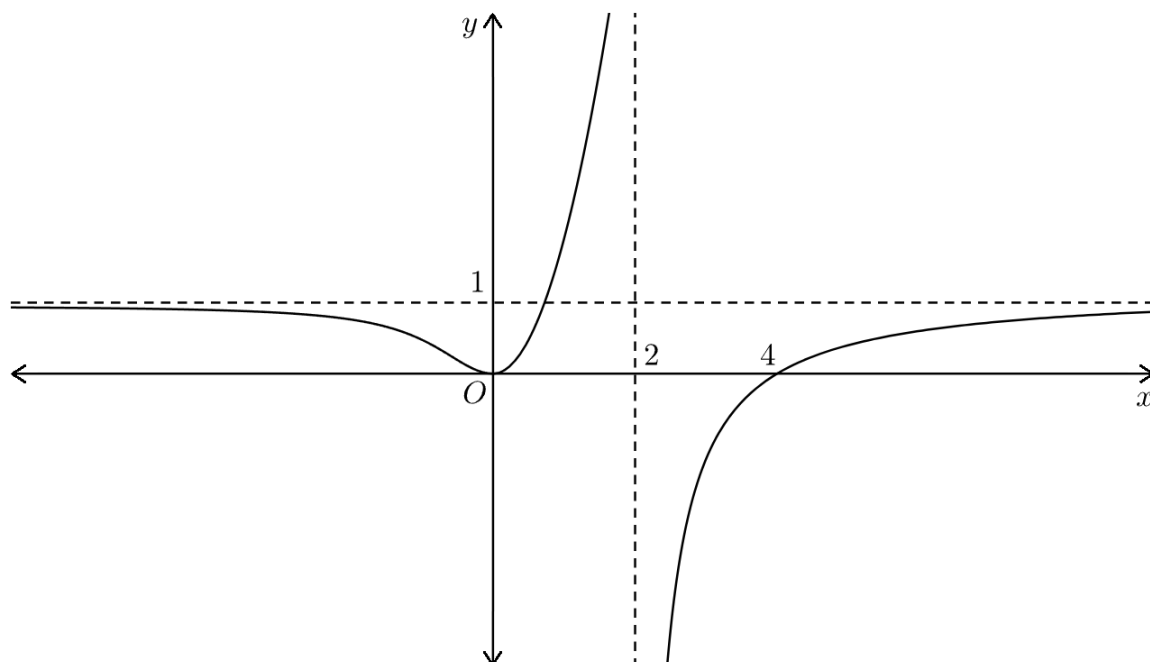
Express your answer as a percentage, to the nearest percent.

Question 12 continues on page 11

Question 12 (continued)

(d) A graph of $y = f(x)$ is shown.

3



On the sheet provided, sketch $y^2 = \frac{1}{f(x)}$.

(e) For non-zero values of p and q , the polynomial equation $P(x) = x^4 - 2px - q = 0$ has 3 distinct roots. Prove that $27p^4 + 16q^3 = 0$.

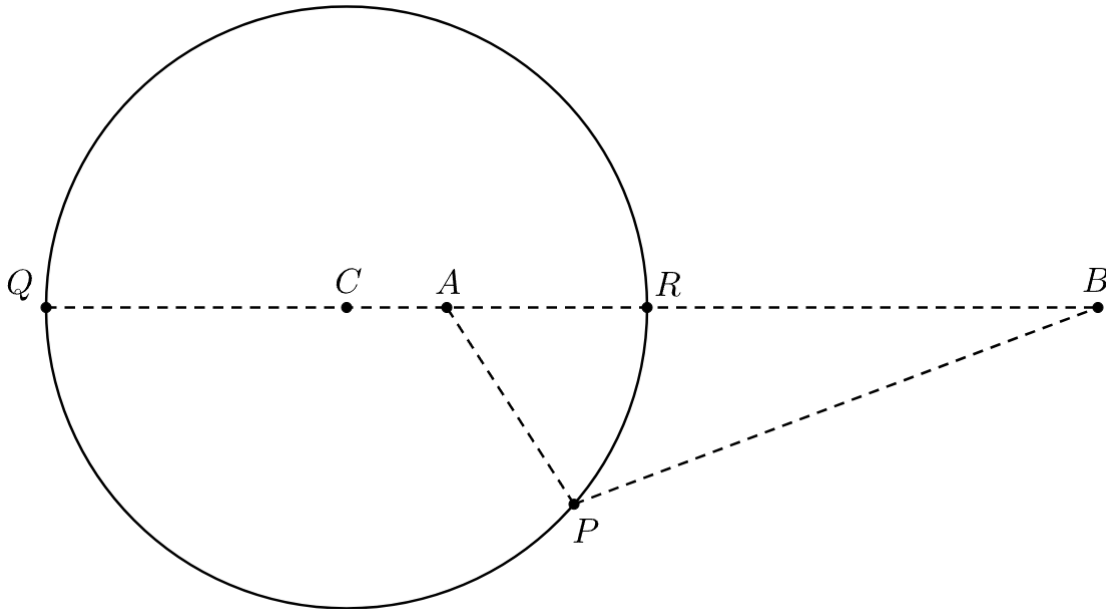
3

End of Question 12

Question 13 (14 marks)

Use a SEPARATE writing booklet.

- (a) (i) Let
- A
- and
- B
- represent the vectors
- $\underline{a}$
- and
- $\underline{b}$
- respectively.

3A point P satisfies $3\vec{PA} = \vec{PB}$, and all possible points P lie on the circle shown.By using vector methods, prove the centre of the circle C has position $\vec{OC} = \frac{9}{8}\underline{a} - \frac{1}{8}\underline{b}$.

- (ii) Find the radius of the circle, in terms of
- $\underline{a}$
- and
- $\underline{b}$
- .

1

- (b) (i) Show that there are
- $\binom{15}{5}$
- ways to distribute 10 identical pineapples between 6 people.

1

- (ii) Hence, by considering the number of ways that 5 people can share up to 10 identical pineapples, prove using a combinatorial argument that

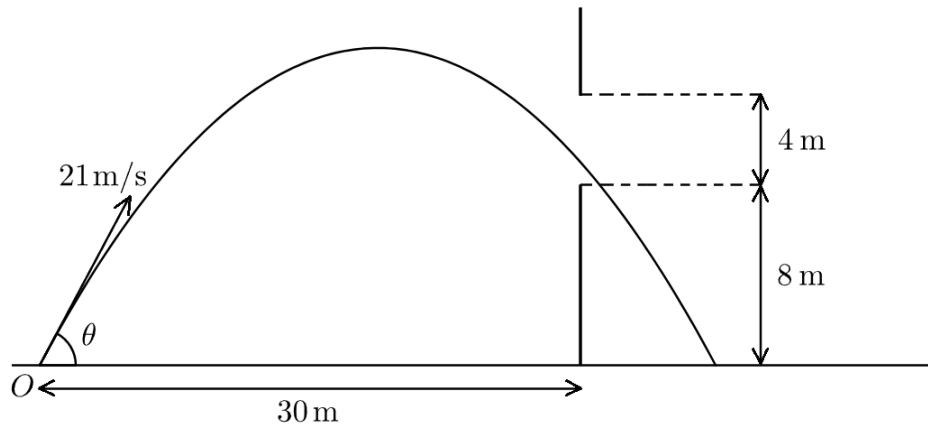
2

$$\sum_{n=4}^{14} \binom{n}{4} = \binom{15}{5}.$$

Question 13 continues on page 13

Question 13 (continued)

- (c) A stream of water is shot from a nozzle from ground level at 21 m/s, at an acute angle of elevation θ . A board is placed vertically 30 metres horizontally from the nozzle, with a 4 metre hole cut such that its base is 8 metres above ground.



You may assume that the position of a particle launched from the origin with speed U and angle ϕ to the horizontal is given by

$$\mathbf{r}(t) = (Ut \cos \phi)\mathbf{i} + \left(Ut \sin \phi - \frac{1}{2}gt^2\right)\mathbf{j} \quad (\text{Do NOT prove this.})$$

where g is the acceleration due to gravity. Take $g = 9.8 \text{ m/s}^2$.

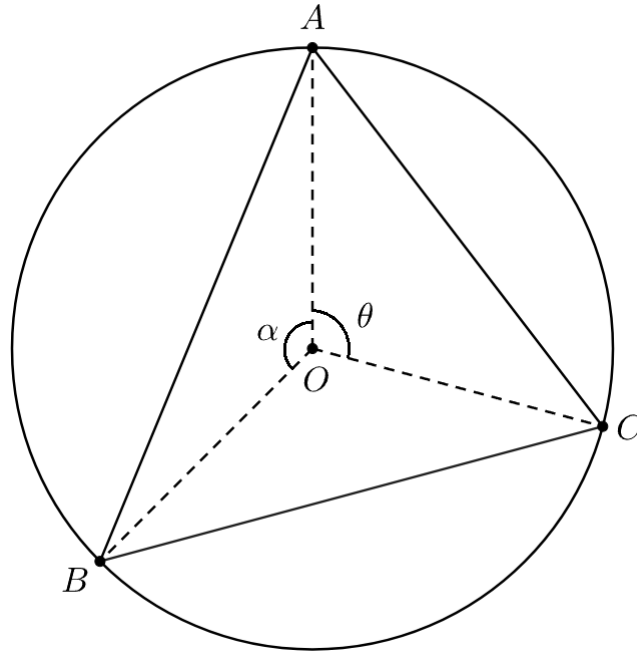
- (i) Prove the Cartesian trajectory of the water follows $y = x \tan \theta - \frac{1}{90}x^2 \sec^2 \theta$. **2**
- (ii) For what range of values of θ will the water stream pass through the hole? **3**
Leave your answer to 2 decimal places.
- (iii) What is the total length of wet ground to the right of the board as θ varies? **2**
Give your answer to the nearest metre.

End of Question 13

Question 14 (17 marks)

Use a SEPARATE writing booklet.

- (a) Let $\triangle ABC$ be inscribed in a circle of radius r centred at O .



Let $\angle AOB = \alpha$ and $\angle AOC = \theta$.

- (i) Prove that the area of triangle ABC is given by

3

$$A = 2r^2 \sin\left(\frac{\alpha + \theta}{2}\right) \sin\frac{\alpha}{2} \sin\frac{\theta}{2}.$$

- (ii) Hence, or otherwise, find the value of θ that maximises the area of the triangle if α is fixed.

2

- (iii) Hence, by treating θ as a fixed angle or otherwise, find the values of α and θ where the maximum area occurs and state the type of triangle formed.

2

Question 14 continues on page 15

Question 14 (continued)

- (b) There is a chemistry course at the University of Old North Scotland that has a course size of 300 students. Students cancel their enrolment right before the course starts which leads to empty spots. Therefore, the lecturer allows for more than 300 enrolments before the course to account for this, given that the lecturer expects to be overbooked at most 1% of the time. What is the range of values of the probability that a singular student comes to class if the lecturer opens up 320 enrolments? 4

You may assume that the sample size is sufficient for the normal approximation to the binomial distribution to be used (without continuity correction).

You are given that if $Z \sim N(0, 1)$ then $P(Z < 1.285) = 0.9$ and $P(Z < 2.326) = 0.99$.

- (c) (i) Prove that if $\dot{w} = lw$, then $w = Ae^{lt}$ for some real numbers A and l . 1
- (ii) Suppose that y_1 and y_2 are the solutions to the paired differential equations 1

$$\dot{y}_1 = 2y_1 + 6y_2,$$

$$\dot{y}_2 = 2y_1 + 3y_2$$

with the initial conditions $y_1(0) = 2$ and $y_2(0) = -3$.

Prove that $\dot{y}_1 + k\dot{y}_2 = (2 + 2k)y_1 + (6 + 3k)y_2$ for all real constants k .

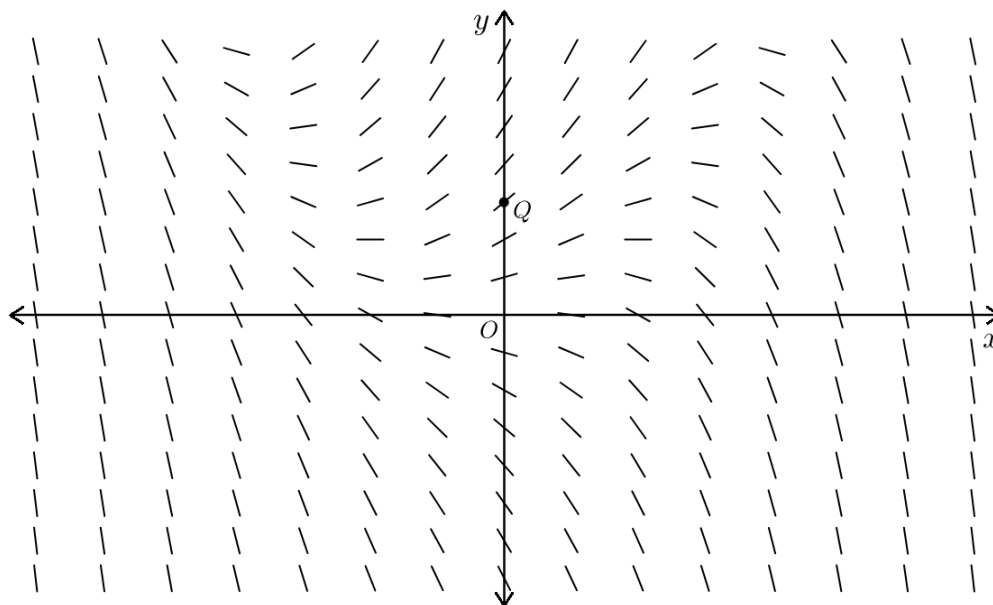
- (iii) Hence, by letting $k = 2$ and another suitable value, solve the differential equations for y_1 and y_2 . 4

End of Paper

Question 12 (a) Place inside booklet for Q12 at the end of the exam

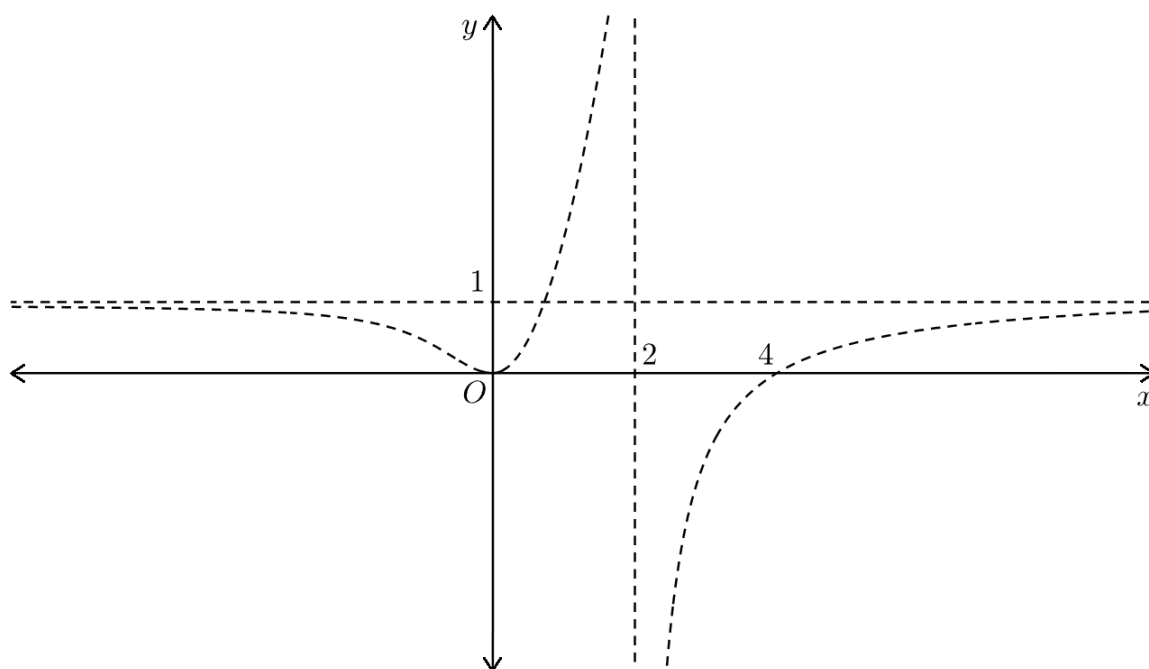
A section of the differential equation $\frac{dy}{dx} = y - x^2$ is shown, along with a point Q .

Sketch a possible solution curve that passes through the point Q .

**Question 12 (d)**

A graph of $y = f(x)$ is shown.

Sketch $y^2 = \frac{1}{f(x)}$.





**SYDNEY
BOYS
HIGH
SCHOOL**

2026

YEAR 12

HSC TASK 4

Mathematics Extension 1

Sample Solutions

NOTE:

This process of checking your mark is about reading the solutions and the comments.

Before putting in an appeal re marking, first consider that the mark is not linked to the amount of writing you have done.

Just because you have shown some 'working' does not justify that your solution is worth any marks.

Students who used pencil, an erasable pen and/or whiteout, may NOT be able to appeal.

MC Answers

1	2	3	4	5	6	7	8	9	10
A	D	D	A	B	C	B	D	C	A

(a) Solve $\frac{2x+1}{3-x} \geq x$.

2

Method ①: critical points (easiest)

$$\frac{2x+1}{3-x} = x, x \neq 3 \text{ (crit. pt. 1)}$$

$$2x+1 = 3x-x^2$$

$$x^2 - x + 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{1 \pm \sqrt{1-4}}{2}$$

= no. solns.

$\therefore$ only $x=3$ is critical



$x=0$ is true $\longrightarrow \frac{2(0)+1}{3-0} \geq 0 \checkmark$

$x=4$ is false: $\frac{2(4)+1}{\cancel{3}-4} \geq 4 \times$

$\therefore x < 3$

Method ②: $x(\text{den})^2$

$$(3-x)^2 \times \frac{2x+1}{3-x} \geq x \times (3-x)^2$$

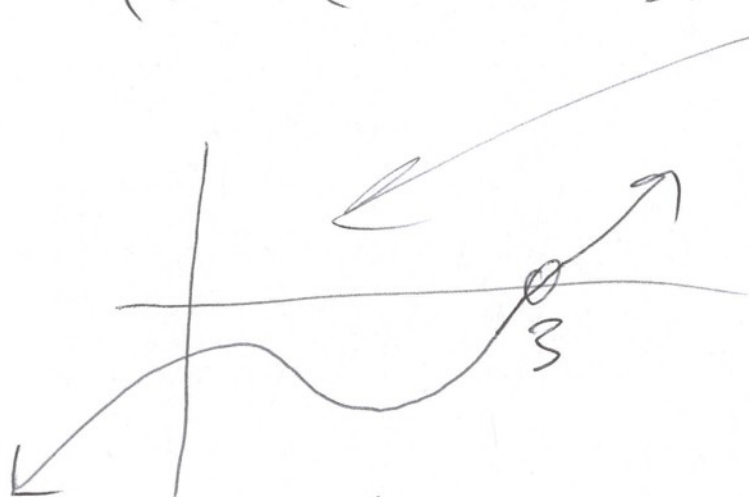
$$(3-x)(2x+1) \geq x(3-x)^2$$

$$(3-x)(2x+1) - x(3-x)^2 \geq 0$$

$$(3-x)(2x+1-x(3-x)) \geq 0$$

$$(3-x)(x^2-x+1) \geq 0 \text{ or } (x-3)(x^2-x+1) \leq 0$$

Same as method ①
no soln.



we want $\leq 0 \rightarrow$ below x -axis, $x \neq 3$
ie $x < 3$

Comments: mostly well done but mistakes tend to happen more with method 2.

- Some students miraculously found solutions that didn't exist for the quadratic component

(b) Let α, β, γ be the roots of the polynomial $3x^3 - 3x^2 + 4x + 9 = 0$.

2

Evaluate $\alpha^2 + \beta^2 + \gamma^2$.

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) \text{ memorise this!} \\ &= \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) \\ &= \left(\frac{3}{3}\right)^2 - 2\left(\frac{4}{3}\right) \\ &= 1 - \frac{8}{3} \\ &= -\frac{5}{3}\end{aligned}$$

Comments:

- well done except for students who got the original identity wrong.

(c) Evaluate the integral $\int_0^{\frac{\pi}{2}} \sin \theta \sin^2 2\theta \, d\theta$.

3

$$= \int_0^{\frac{\pi}{2}} \sin \theta (2 \sin \theta \cos \theta)^2 \, d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} \sin^3 \theta \cos^2 \theta \, d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} \sin \theta (1 - \cos^2 \theta) \cos^2 \theta \, d\theta$$

because we want to use $\int f'(x) f(x)^n \, dx = \left[\frac{f(x)^{n+1}}{n+1} \right]$

$$= 4 \int_0^{\frac{\pi}{2}} \sin \theta \cos^2 \theta - \sin \theta \cos^4 \theta \, d\theta$$

$$= -4 \int_0^{\frac{\pi}{2}} \overset{f'(x)}{\sin \theta} \overset{f(x)^2}{\cos^2 \theta} - \overset{f'(x)}{\sin \theta} \overset{f(x)^4}{\cos^4 \theta} \, d\theta$$

$$= -4 \left[\frac{\cos^3 \theta}{3} - \frac{\cos^5 \theta}{5} \right]_0^{\frac{\pi}{2}}$$

$$= -4 \left(\left(\frac{\cos^3 \frac{\pi}{2}}{3} - \frac{\cos^5 \frac{\pi}{2}}{5} \right) - \left(\frac{\cos^3 0}{3} - \frac{\cos^5 0}{5} \right) \right)$$

$$= -4 \left(0 - \left(\frac{1}{3} - \frac{1}{5} \right) \right)$$

$$= \frac{8}{15}$$

Question 11 (c) continued on page #5

method ②: far far harder, $\therefore$ lot more mistakes

$$\begin{aligned}
 & \int_0^{\frac{\pi}{2}} \sin \theta \cdot \frac{1}{2}(1 - \cos 4\theta) d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin \theta - \sin \theta \cos 4\theta d\theta \quad \sin A \cos B = \frac{1}{2}(\sin(A+B) + \sin(A-B)) \\
 &= -\frac{1}{2}(\sin 5\theta + \sin(-3\theta)) \quad A = \theta, B = 4\theta \\
 &= -\frac{1}{2}(\sin 5\theta - \sin 3\theta) \\
 &= \frac{1}{2} \left[\cos \theta - \frac{1}{2} \cdot \frac{\cos 5\theta}{-5} + \frac{1}{2} \frac{\cos 3\theta}{-3} \right]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{2} \left((-\cos \frac{\pi}{2} + \frac{1}{10} \cos \frac{5\pi}{2} - \frac{1}{6} \cos \frac{3\pi}{2}) - (-\cos 0 + \frac{1}{10} \cos 0 - \frac{1}{6} \cos 0) \right) \\
 &= \frac{1}{2} \left((0 + 0 - 0) - (-1 + \frac{1}{10} - \frac{1}{6}) \right) \\
 &= \frac{1}{2} \left(-(-\frac{16}{15}) \right) \\
 &= \frac{8}{15}
 \end{aligned}$$

Comments:

- Some students chose to transform $\cos^2 \theta = 1 - \sin^2 \theta$. This very quickly led to a dead-end which they were, unfortunately, unable to integrate.
- Method 2 has by far the more confusing (-) signs, which caused mistakes & was the more common method.

(d) Find the coefficient of x in the expansion of $(1 - 3x)(2x - 1)^9$.

2

$$(1-3x)((2x)^0(-1)^9{}^9C_0 + \underbrace{{}^9C_1(-1)^8(2x)^1}_{\text{coeff. of } x} + \dots)$$

We want coeff. of $x \rightarrow$

$$-3x \times \text{constant} + 1x$$

$$= -3 \times (-1)^9 \times {}^9C_0 + {}^9C_1(-1)^8(2)^1$$

$$= 3 \times 1 + 9 \times 2$$

$$= 21$$

Comments:

- there were some ridiculously large numbers which were clearly wrong.
- a few students forgot the combinatorics component ${}^nC_r = {}^nC_0$ & nC_1 here in this Q. You might want to brush up on it.

(e) Prove that $\tan\left(\frac{\pi}{4} + \theta\right) = \sec 2\theta + \tan 2\theta$ by using the substitution $t = \tan \theta$.

3

using the double angle on t -formula $\theta = 2\theta$
gives these: $\sin 2\theta = \frac{2t}{1+t^2}$, $\cos 2\theta = \frac{1-t^2}{1+t^2}$, $\tan 2\theta = \frac{2t}{1-t^2}$

$$\text{LHS} = \tan\left(\frac{\pi}{4} + \theta\right) \quad \text{compound angle formula}$$

$$= \frac{\tan \frac{\pi}{4} + \tan \theta}{1 - \tan \frac{\pi}{4} \tan \theta}$$

$$= \frac{1 + \tan \theta}{1 - \tan \theta} \quad \tan \frac{\pi}{4} = 1$$

$$= \frac{1+t}{1-t}$$

$$\text{RHS} = \sec 2\theta + \tan 2\theta$$

$$= \frac{1}{\cos 2\theta} + \frac{2t}{1-t^2}$$

$$= \frac{1+t^2+2t}{1-t^2}$$

$$= \frac{(1+t)^2}{(1-t)(1+t)}$$

$$= \text{LHS}$$

Comments:

This Q. was really well done

(f) It is given that a and b are real numbers such that $\sin^{-1} a = \cos^{-1} b = \theta$

2

Prove that (a, b) lie on the unit circle in the first quadrant.

$$\theta = \sin^{-1}(a) \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\theta = \cos^{-1}(b) \in [0, \pi]$$

θ must be between 0 & $\frac{\pi}{2}$ (common range),
ie Quadrant I proven.

$$a = \sin \theta$$

$$b = \cos \theta$$

Unit circle $x^2 + y^2 = 1 \Leftrightarrow a^2 + b^2 = 1$
Pythagorean identity $a^2 + b^2 = \sin^2 \theta + \cos^2 \theta$
 $= 1$ is true

$\therefore$ proven unit circle.

Comments:

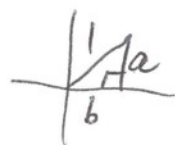
— The single most common error was not answering both parts:

1 mark for unit circle

1 mark for first quadrant.

— A lot of students baldly stated that $a > 0, b > 0$ because they had a triangle (in Q.1)

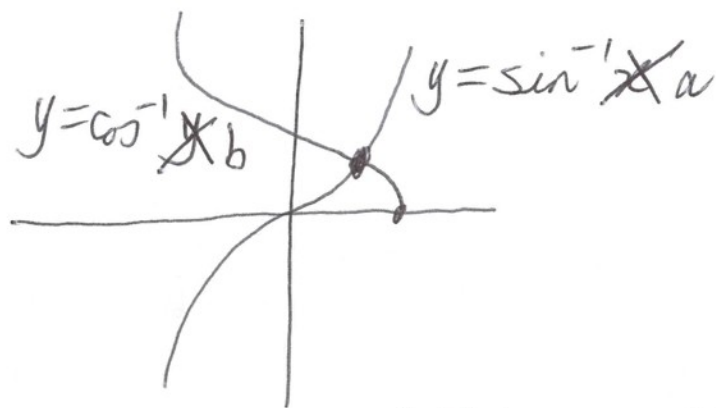
This is not valid.



— P.T.O. to see more proofs.

Question 11 (f) continues on page #9

Question 11 (f) (continued)



$$a = b = \frac{1}{\sqrt{2}} \rightarrow \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = 1$$

$\therefore$ first quadrant of unit circle $a^2 + b^2 = 1$

OR

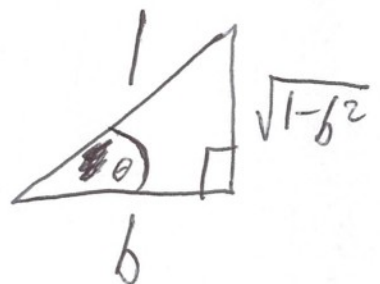
$$\sin^{-1}a = \cos^{-1}b$$

$$\sin(\sin^{-1}a) = \sin(\cos^{-1}b)$$

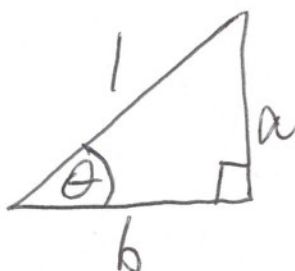
$$a = \sqrt{1-b^2}$$

$$a^2 = 1-b^2$$

$$a^2 + b^2 = 1$$



OR



$$\theta = \sin^{-1}a = \cos^{-1}b$$

$$\sin \theta = \frac{a}{1}$$

$$\cos \theta = \frac{b}{1}$$

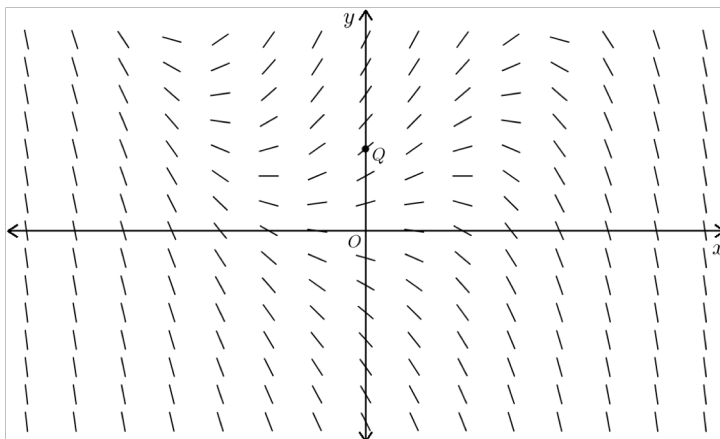
$$\therefore a^2 + b^2 = 1 \text{ (P.T.)}$$

End of Q11 Solutions

- (a) A section of the differential equation $\frac{dy}{dx} = y - x^2$ is shown, along with a point Q .

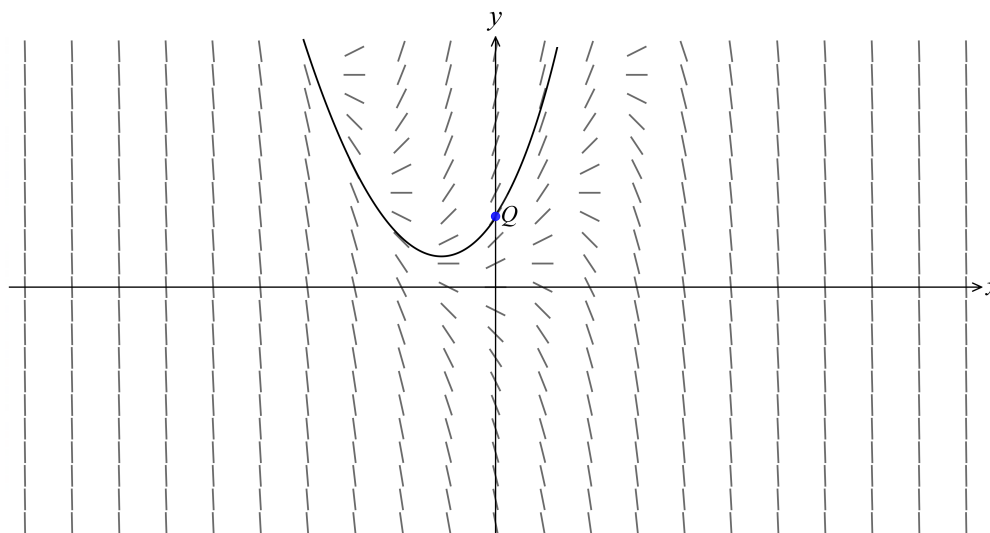
2

w

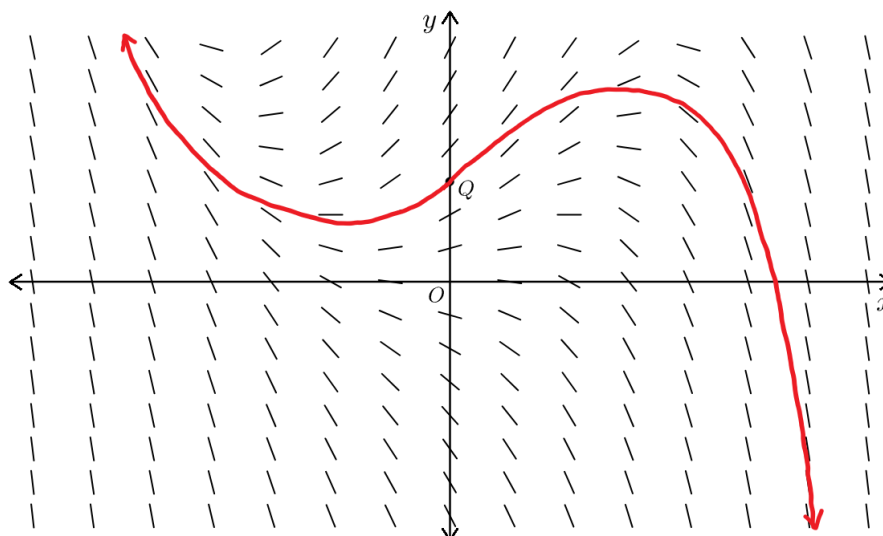


On the sheet provided, sketch a possible solution curve that passes through the point Q .

Possible Answer 1



Possible Answer 2



(a) (continued)

Well done.

No half marks

To receive full marks, the curve needed to

- *be smooth and have the correct overall shape*
- *not cross any of the segments*
- *follow the correct directions indicated by the slope field*
- *pass through the point Q*

Common errors:

- *Sketching only the left-hand or right-hand portion of the curve. No marks were awarded for an incomplete curve.*
- *Difficulty following the directions of the slope field, particularly on the right-hand side.*
- *Drawing a curve that crossed one or more of the given segments.*

(b) Let n be an odd positive integer.

Prove by mathematical induction that $5^n + 2^n$ is divisible by 7.

Base case $n = 1$:

$$5^1 + 2^1 = 7 \text{ which is divisible by 7}$$

Therefore true for $n = 1$.

Assume true for $n = 2k + 1$, i.e. $5^{2k+1} + 2^{2k+1} = 7M, k \in \mathbb{Z}^+, M \in \mathbb{Z}^+$

Prove true for $n = 2k + 3$, i.e. $5^{2k+3} + 2^{2k+3} = 7N, k \in \mathbb{Z}^+, N \in \mathbb{Z}^+$

$$\begin{aligned} \text{LHS} &= 5^{2k+3} + 2^{2k+3} \\ &= 5^2 \times 5^{2k+1} + 2^2 \times 2^{2k+1} \\ &= 25(7M - 2^{2k+1}) + 4 \times 2^{2k+1} \\ &= 7 \times 25M - 25 \times 2^{2k+1} + 4 \times 2^{2k+1} \\ &= 7 \times 25M - 21 \times 2^{2k+1} \\ &= 7(25M - 3 \times 2^{2k+1}) \\ &= 7N \quad (\text{where } N = 25M - 3 \times 2^{2k+1} \text{ which is an integer}) \end{aligned}$$

$7N$ is divisible by 7, as required.

Hence, by mathematical induction, $5^n + 2^n$ is divisible by 7, for all n be an odd positive integer.

Well done.

Most students were able to follow the correct proof structure of mathematical induction.

As this was a proof question, errors in establishing the base case or stating the induction assumption affected students' ability to complete the proof successfully.

Common mistakes:

- *Students did not realise that n is odd, which prevented them from completing the proof successfully.*
- *Students did not follow the standard structure of a proof by mathematical induction. Please refer to the 2025 and 2024 Mathematics Extension 1 HSC papers for examples of the expected proof structure.*
- *A few students made careless errors, such as miswriting 2 as 7, which led to incorrect solutions. Students should take care to check their working and ensure that numbers and expressions are copied accurately.*

(c) Some butter is in the shape of a sphere, with radius r .

It is placed into an oven so that its volume is decreasing at a rate proportional to the square root of its surface area, S . You may assume as the butter melts, it is melting uniformly along its surface.

(i) Show that $\frac{dr}{dt} = \frac{-k}{\sqrt{S}}$ for some positive constant $k > 0$.

1

$$S = 4\pi r^2 \text{ (area of a sphere)}$$

$$\frac{dV}{dt} = -k\sqrt{S} \text{ (given)}$$

$$\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{S} \times (-k\sqrt{S})$$

$$= \frac{-k}{\sqrt{S}}$$

Generally well done.

No half marks.

As this is a show type question, sufficient working was required to demonstrate the necessary steps and justify the answer.

(c) (continued)

- (ii) The butter initially had radius 2 cm. By how much does the total melting time exceed the time needed for the butter to shrink to half of its original radius? **3**

Express your answer as a percentage, to the nearest percent.

$$\frac{dr}{dt} = -\frac{k}{\sqrt{S}} = -\frac{k}{\sqrt{4\pi r^2}} = -\frac{k}{2\sqrt{\pi}r}$$

$$\therefore 2\sqrt{\pi}r dr = -k dt$$

$$\int_2^r 2\sqrt{\pi}r dr = -\int_0^t k dt$$

$$\therefore \left[\sqrt{\pi}r^2 \right]_2^r = -[kt]_0^t$$

$$\therefore \sqrt{\pi}r^2 - 4\sqrt{\pi} = -kt$$

$$\therefore t = \frac{4\sqrt{\pi} - \sqrt{\pi}r^2}{k}$$

$$t(1) = \frac{3\sqrt{\pi}}{k}$$

$$t(0) = \frac{4\sqrt{\pi}}{k}$$

$$\begin{aligned} \frac{t(0) - t(1)}{t(1)} \times 100\% &= \frac{\frac{4\sqrt{\pi}}{k} - \frac{3\sqrt{\pi}}{k}}{\frac{3\sqrt{\pi}}{k}} \times 100\% \\ &= \frac{1}{3} \times 100\% \end{aligned}$$

B 33%

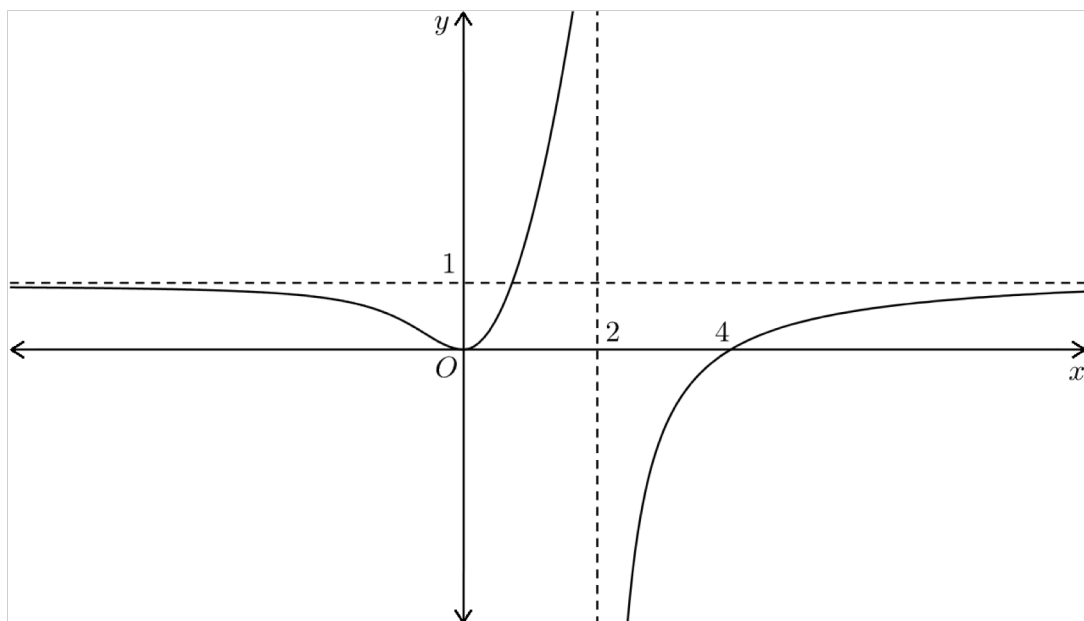
Generally well done.

Common mistakes:

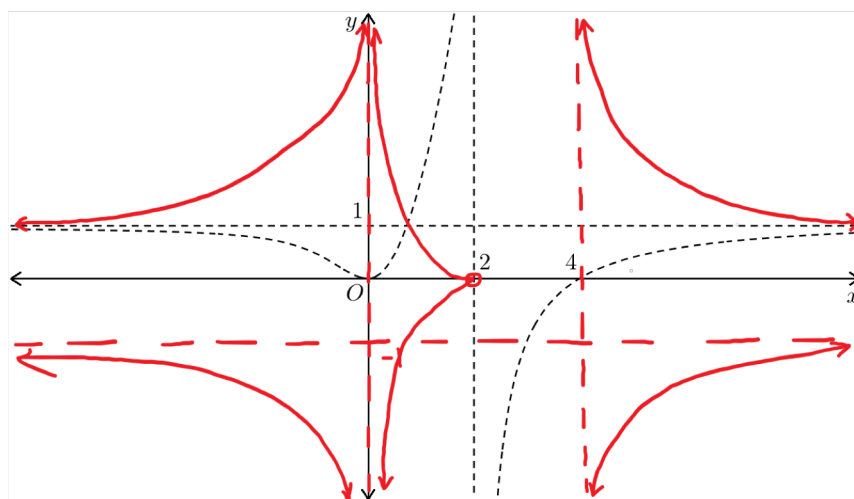
- Some students had difficulty separating the variables correctly that they did not substitute $4\pi r^2$ for S , which prevented them from integrating correctly to obtain the required relationship between r and t .
- Some students did not recognise that $\sqrt{4\pi} = 2\sqrt{\pi}$.
No marks were deducted. However, being able to simplify expressions is essential for solving problems efficiently and accurately.
- Some students did not read the question carefully. The most common incorrect answer was 75%, which suggests that the final question requirement was not fully considered.

(d) A graph of $y = f(x)$ is shown.

3



On the sheet provided, sketch $y^2 = \frac{1}{f(x)}$.



Generally well done.

To achieve full marks, the curve needed to:

- *have the correct general shape*
- *clearly show the asymptotes: $y = \pm 1$, y -axis and $x = 4$*
- *intersect the original curve at $y = \pm 1$*
- *be symmetrical*
- *have no graph between $x = 2$ to $x = 4$*

Common mistakes:

- *Some students did not recognize that $y = -1$ is an asymptote as well.*
- *Some students forgot to erase the section of the graph between $x = 2$ to $x = 4$, resulting in an incorrect sketch.*

Students should check their final graph carefully and ensure that only the required sections are shown.

(e) For non-zero values of p and q , the polynomial equation $P(x) = x^4 - 2px - q = 0$ has 3 distinct roots. Prove that $27p^4 + 16q^3 = 0$.

3

Method 1a:

$$P'(x) = 4x^3 - 2p = 0$$

$$\therefore x^3 = \frac{p}{2}$$

$$\begin{aligned} P(x) &= x^4 - 2px - q \\ &= x(x^3 - 2p) - q \\ &= x\left(\frac{p}{2} - 2p\right) - q \\ &= -\frac{3}{2}px - q \\ &= 0 \end{aligned}$$

$$\therefore x = -\frac{2q}{3p}$$

$$\therefore \left(-\frac{2q}{3p}\right)^3 = \frac{p}{2}$$

$$-\frac{8q^3}{27p^3} = \frac{p}{2}$$

$$\therefore -16q^3 = 27p^4$$

$$\therefore 27p^4 + 16q^3 = 0$$

Method 1b:

$$P'(x) = 4x^3 - 2p = 0$$

$$\therefore p = 2x^3$$

$$\begin{aligned} P(x) &= x^4 - 2px - q \\ &= x^4 - 2 \times 2x^3 \times x - q \\ &= -3x^4 - q \\ &= 0 \end{aligned}$$

$$\therefore q = -3x^4$$

$$\begin{aligned} \therefore 27p^4 + 16q^3 &= 27(2x^3)^4 + 16(-3x^4)^3 \\ &= 27 \times 16x^{12} + 16 \times (-27)x^{12} \\ &= 0 \end{aligned}$$

Method 1c:

$$P'(x) = 4x^3 - 2p = 0$$

$$\therefore x = \left(\frac{p}{2}\right)^{\frac{1}{3}}$$

$$\begin{aligned} P(x) &= \left(\frac{p}{2}\right)^{\frac{4}{3}} - 2p\left(\frac{p}{2}\right)^{\frac{1}{3}} - q \\ &= \left(\frac{p}{2}\right)^{\frac{1}{3}}\left(\frac{p}{2} - 2p\right) - q \\ &= \left(\frac{p}{2}\right)^{\frac{1}{3}}\left(-\frac{3p}{2}\right) - q \\ &= -\frac{3p}{2}\left(\frac{p}{2}\right)^{\frac{1}{3}} - q \\ &= 0 \end{aligned}$$

$$i.e. 3p^{\frac{4}{3}} = -2^{\frac{4}{3}}q$$

$$\therefore \left(3p^{\frac{4}{3}}\right)^3 = \left(-2^{\frac{4}{3}}q\right)^3$$

$$\therefore 27p^4 = -16q^3$$

$$\therefore 27p^4 + 16q^3 = 0$$

(e) (continued)

Method 2:

Let $\alpha, \alpha, \beta, \gamma$ be the roots of $P(x) = 0$

$$2\alpha + \beta + \gamma = 0 \quad \textcircled{1}$$

$$\alpha^2 + 2\alpha\beta + 2\alpha\gamma + \beta\gamma = 0 \quad \textcircled{2}$$

$$\alpha^2\beta + \alpha^2\gamma + 2\alpha\beta\gamma = 2p \quad \textcircled{3}$$

$$\alpha^2\beta\gamma = -q \quad \textcircled{4}$$

$$\begin{aligned} -\beta\gamma &= \alpha^2 + 2\alpha\beta + 2\alpha\gamma \\ &= \alpha^2 + 2\alpha(\beta + \gamma) \\ &= \alpha^2 + 2\alpha(-2\alpha) \\ &= -3\alpha^2 \end{aligned}$$

$$\therefore \beta\gamma = 3\alpha^2$$

$$\begin{aligned} 27p^4 + 16q^3 &= \frac{27}{16}(2p)^4 + 16q^3 \\ &= \frac{27}{16}(\alpha^2\beta + \alpha^2\gamma + 2\alpha\beta\gamma)^4 + 16(-\alpha^2\beta\gamma)^3 \\ &= \frac{27}{16}[\alpha^2(\beta + \gamma) + 2\alpha(3\alpha^2)]^4 - 16\alpha^6(3\alpha^2)^3 \\ &= \frac{27}{16}[\alpha^2(-2\alpha) + 2\alpha(3\alpha^2)]^4 - 16\alpha^6(3\alpha^2)^3 \\ &= \frac{27}{16}(-2\alpha^3 + 6\alpha^3)^4 - 16 \times 27\alpha^{12} \\ &= \frac{27}{16}(4\alpha^3)^4 - 16 \times 27\alpha^{12} \\ &= 27 \times 16\alpha^{12} - 16 \times 27\alpha^{12} \\ &= 0 \end{aligned}$$

Generally well done.

Method 2 was the more popular approach; however, no students were able to apply it successfully to complete the question.

To receive the first mark, all four sums and products of roots were required to be established.

Using multiplicity was the recommended approach. However, some students experienced difficulty manipulating the algebraic expressions required to complete the proof.

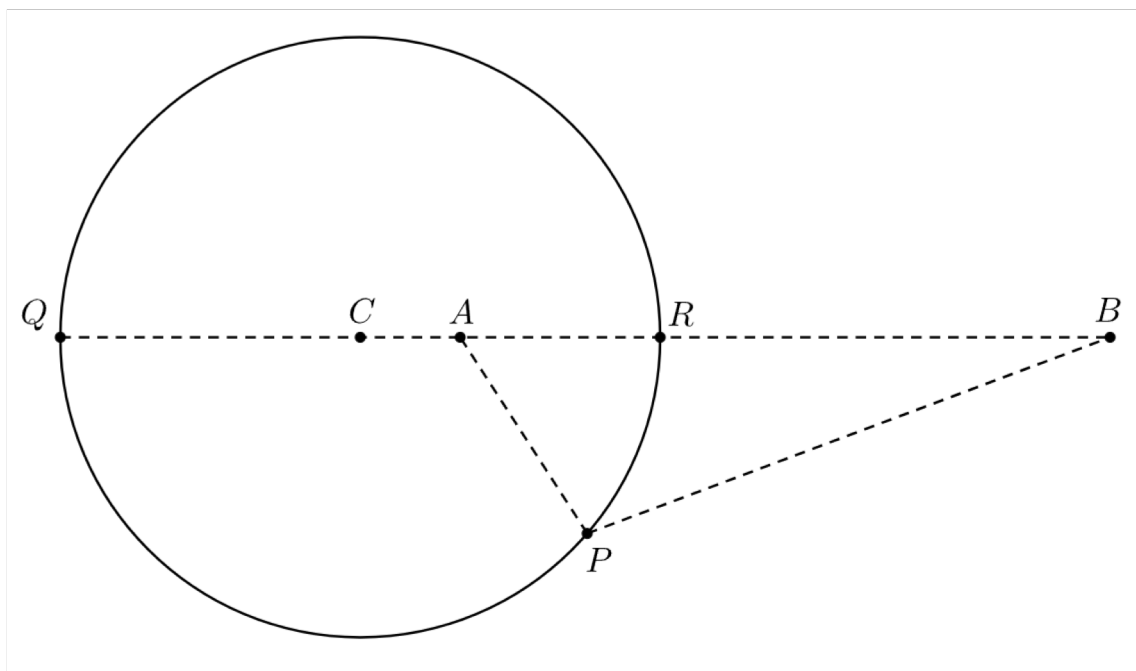
Students should continue to develop their algebraic manipulation skills, as these are essential for constructing clear and accurate proofs.

End of Q12 Solutions

- (a) (i) Let A and B represent the vectors $\underline{a}$ and $\underline{b}$ respectively.

3

A point P satisfies $3\vec{PA} = \vec{PB}$, and all possible points P lie on the circle shown.



By using vector methods, prove the centre of the circle C has position $\vec{OC} = \frac{9}{8}\underline{a} - \frac{1}{8}\underline{b}$.

Since C is the centre of circle and Q, C and R are collinear, QR is diameter.

$\therefore C$ is a midpoint of QR

Q and R satisfy the condition defining P i.e

$$P = Q, \quad 3\vec{QA} = \vec{QB}$$

$$P = R, \quad 3\vec{RA} = \vec{RB}$$

$$\text{Let } \vec{OA} = \underline{a}, \quad \vec{OB} = \underline{b}$$

$$\therefore \vec{QA} = \frac{1}{3}\vec{QB} \quad (1)$$

$$\vec{QA} = \vec{OA} - \vec{OQ}$$

$$= \underline{a} - \vec{OQ} \quad (2)$$

$$\vec{QB} = \vec{OB} - \vec{OQ}$$

$$= \underline{b} - \vec{OQ} \quad (3)$$

Sub (2) and (3) into (1)

$$\underline{a} - \vec{OQ} = \frac{1}{3}(\underline{b} - \vec{OQ})$$

$$3\underline{a} - 3\vec{OQ} = \underline{b} - \vec{OQ}$$

$$2\vec{OQ} = 3\underline{a} - \underline{b}$$

$$\therefore \vec{OQ} = \frac{1}{2}(3\underline{a} - \underline{b}) \quad (4)$$

(a) (continued)

Since R lies between A and B

$$\vec{RA} = -\frac{1}{3}\vec{RB}$$

$$\vec{OA} - \vec{OR} = -\frac{1}{3}(\vec{OB} - \vec{OR})$$

$$\underline{a} - \vec{OR} = -\frac{1}{3}(\underline{b} - \vec{OR})$$

$$-3\underline{a} + 3\vec{OR} = \underline{b} - \vec{OR}$$

$$4\vec{OR} = 3\underline{a} + \underline{b}$$

$$\therefore \vec{OR} = \frac{1}{4}(3\underline{a} + \underline{b}) \quad (5)$$

$$\therefore \vec{OC} = \frac{1}{2}(\vec{OA} + \vec{OB})$$

$$= \frac{1}{2}\left(\frac{3}{2}\underline{a} - \frac{1}{2}\underline{b} + \frac{3}{4}\underline{a} + \frac{1}{4}\underline{b}\right) \quad (\text{From (4) and (5)})$$

$$= \frac{1}{2}\left(\frac{9}{4}\underline{a} - \frac{1}{4}\underline{b}\right)$$

$$= \frac{9}{8}\underline{a} - \frac{1}{8}\underline{b}$$

Marking Scheme:

1 mark: Correctly finds either position vector $\vec{OR}$ or $\vec{OQ}$.

1 mark: Correctly find the remaining position vector ($\vec{OR}$ or $\vec{OQ}$).

1 mark: Complete the proof using valid vector methods to derive the required result.

Marker's comments:

- This question was poorly answered by majority of candidates.
- The relation $3PA = PB$ specifies scalar distance, making the assumption $3\vec{PA} = \vec{PB}$ a major conceptual error since P, A and B generally form a triangle rather than parallel vectors. Conversely, Q and R are unique boundary points where P is collinear with A and B making them the only locations where this magnitude ratio can be converted into directed vector equations.
- Some candidates wrote correct vector relationships but made no meaningful progress towards the required result and therefore could not be awarded marks.

(a) (continued)

(ii) Find the radius of the circle, in terms of $\underline{a}$ and $\underline{b}$.

1

$$\begin{aligned} r &= \frac{1}{2} |\overrightarrow{QR}| \\ &= \frac{1}{2} |\overrightarrow{OR} - \overrightarrow{OQ}| \\ &= \frac{1}{2} \left| \left(\frac{3}{4}\underline{a} + \frac{1}{4}\underline{b} \right) - \left(\frac{3}{2}\underline{a} - \frac{1}{2}\underline{b} \right) \right| \\ &= \frac{1}{2} \left| -\frac{3}{4}\underline{a} + \frac{3}{4}\underline{b} \right| \\ &= \frac{3}{8} |\underline{b} - \underline{a}| \end{aligned}$$

Marking Scheme:

1 mark: Correct answer.

Marker's Comments:

- This question was poorly answered, with very few candidates obtaining the correct radius.
- Some candidates got close to the correct answer but failed to express the radius as a length not vector.

- (b) (i) Show that there are $\binom{15}{5}$ ways to distribute 10 identical pineapples between 6 people. 1

let 10 identical pineapples be represented by 10 identical symbols, +.

To divide these 10 pineapples among 6 distinct people, we need

$(6-1=5)$ dividers represented by symbol |

Example ++|+++|+|++||++

represents person 1 gets 2 pineapples, person 2 gets 3 pineapples, person 3 gets 1 pineapple, person 4 gets 2 pineapples, person 5 gets 0 pineapples and person 6 gets 2 pineapples.

To order these, it is equal to having 15 +, and choosing any 5 of them to turn to |.

$\therefore \binom{10+5}{5} = \binom{15}{5}$ ways

Marking Scheme:

- 1 mark:** Provides a correct explanation or proof for $\binom{15}{5}$ by explicitly defining the 10 pineapples and 5 dividers within the context of the problem.

Marker's Comment:

- Candidates did not receive the mark if insufficient explanation were given to explicitly link the math to the scenario of the question.

(b) (continued)

- (ii) Hence, by considering the number of ways that 5 people can share up to 10 identical pineapples, prove using a combinatorial argument that 2

$$\sum_{n=4}^{14} \binom{n}{4} = \binom{15}{5}.$$

Consider 5 people sharing and using method in (i):

0 pineapples, with 0 + and 4 |, so there are $\binom{4}{4}$ ways

1 pineapple, with 1 + and 4 |, so there are $\binom{5}{4}$ ways

2 pineapples, with 2 + and 4 |, so there are $\binom{6}{4}$ ways

0
0
0

10 pineapples, with 10 + and 4 |, so there are $\binom{14}{4}$ ways

∴ To share 10 pineapples between 5 people, there are

$$\binom{4}{4} + \binom{5}{4} + \binom{6}{4} + \dots + \binom{14}{4} = \sum_{n=4}^{14} \binom{n}{4} \text{ ways}$$

Any left over in each case (out of the 10) are automatically given to the sixth person. This is an exact correspondence to b) i).

$$\therefore \sum_{n=4}^{14} \binom{n}{4} = \binom{15}{5}$$

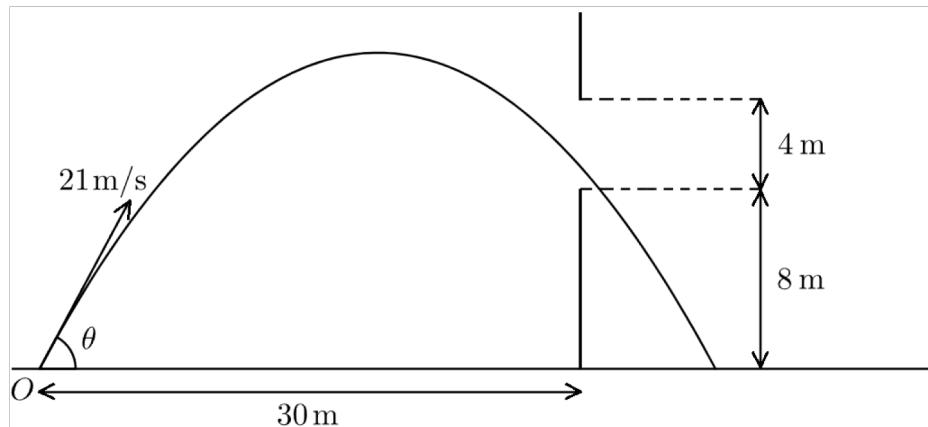
Marking Scheme:

- 1 mark:** Correctly establish the case by case sum for 5 people with brief description.
- 1 mark:** Connect the “leftovers” to the 6th person to link back to the result in part b) i).

Marker's Comments:

- This question was very poorly done.
- Many candidates struggled failed to give "combinatorial argument" and did not know how to break down the problem into mutually exclusive cases based on the number of pineapples distributed.
A combinatorial argument is a proof where you show an identity is true by counting the same group of objects in two different ways with explanation, not expanding factorials.

- (c) A stream of water is shot from a nozzle from ground level at 21 m/s, at an acute angle of elevation θ . A board is placed vertically 30 metres horizontally from the nozzle, with a 4 metre hole cut such that its base is 8 metres above ground.



You may assume that the position of a particle launched from the origin with speed U and angle ϕ to the horizontal is given by

$$\mathbf{r}(t) = (Ut \cos \phi)\mathbf{i} + \left(Ut \sin \phi - \frac{1}{2}gt^2\right)\mathbf{j} \quad (\text{Do NOT prove this.})$$

where g is the acceleration due to gravity. Take $g = 9.8 \text{ m/s}^2$.

- (i) Prove the Cartesian trajectory of the water follows $y = x \tan \theta - \frac{1}{90}x^2 \sec^2 \theta$. 2

$$\begin{aligned} U &= 21 & g &= 9.8 \\ x &= 21t \cos \theta \\ \therefore t &= \frac{x}{21 \cos \theta} & \text{①} \\ y &= 21t \sin \theta - \frac{1}{2} \times 9.8 t^2 \\ &= 21t \sin \theta - 4.9 t^2 & \text{②} \\ \text{Sub ① into ②} \\ y &= 21 \sin \theta \left(\frac{x}{21 \cos \theta} \right) - 4.9 \left(\frac{x}{21 \cos \theta} \right)^2 \\ &= x \tan \theta - \frac{4.9 x^2}{21^2 \cos^2 \theta} \\ \therefore y &= x \tan \theta - \frac{1}{90} x^2 \sec^2 \theta \end{aligned}$$

(c) (i) (continued)

Marking Scheme:

1 mark: Correctly obtains $t = \frac{x}{21 \cos \theta}$ (or $t = \frac{x}{U \cos \theta}$).

1 mark: Correctly substitute t into $y = 21t \sin \theta - 4.9t^2$ (or $y = Ut \sin \theta - 4.9t^2$) to obtain the required result.

Marker's comment:

- This question was well done by the majority of candidates.

(c) (continued)

(ii) For what range of values of θ will the water stream pass through the hole?

3

Leave your answer to 2 decimal places.

$$x = 30$$

$$8 < y < 12$$

$$\text{From (i)} \quad y = 30 \tan \theta - \frac{1}{2} \times 30^2 \sec^2 \theta$$

$$= 30 \tan \theta - 10 \sec^2 \theta$$

$$= 30 \tan \theta - 10(1 + \tan^2 \theta)$$

$$\therefore y = -10 \tan^2 \theta + 30 \tan \theta - 10$$

Case 1: $-10 \tan^2 \theta + 30 \tan \theta - 10 > 8$

$$10 \tan^2 \theta - 30 \tan \theta + 10 < -8$$

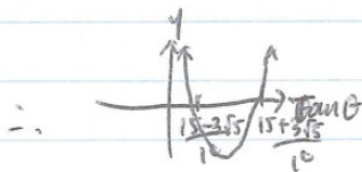
$$10 \tan^2 \theta - 30 \tan \theta + 18 < 0$$

$$5 \tan^2 \theta - 15 \tan \theta + 9 < 0$$

$$\text{When } 5 \tan^2 \theta - 15 \tan \theta + 9 = 0$$

$$\tan \theta = \frac{15 \pm \sqrt{(-15)^2 - 4 \times 5 \times 9}}{10}$$

$$= \frac{15 \pm 3\sqrt{5}}{10}$$



$$\therefore \frac{15-3\sqrt{5}}{10} < \tan \theta < \frac{15+3\sqrt{5}}{10} \quad (\text{From sketch above})$$

$$\therefore 39.66^\circ < \theta < 65.27^\circ$$

(c) (ii) (continued)

Case 2 $-10 \tan^2 \theta + 30 \tan \theta - 10 < 12$

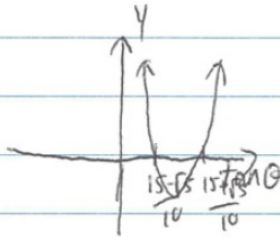
$$10 \tan^2 \theta - 30 \tan \theta + 10 > -12$$

$$10 \tan^2 \theta - 30 \tan \theta + 22 > 0$$

$$5 \tan^2 \theta - 15 \tan \theta + 11 > 0$$

When $5 \tan^2 \theta - 15 \tan \theta + 11 = 0$

$$\tan \theta = \frac{15 \pm \sqrt{(-15)^2 - 4 \times 5 \times 11}}{10}$$



$$= \frac{15 \pm \sqrt{5}}{10}$$

$$\therefore \tan \theta < \frac{15 - \sqrt{5}}{10} \quad \text{or} \quad \tan \theta > \frac{15 + \sqrt{5}}{10}$$

$$\therefore \theta < 51.92^\circ \quad \text{or} \quad \theta > 59.88^\circ$$

$\therefore$ Solution = The intersection of solution of the 2 cases

$$\therefore 39.66^\circ < \theta < 51.92^\circ \quad \text{or} \quad 59.88^\circ < \theta < 65.27^\circ$$

Marking Scheme:

- 1 mark:** Correct working to establish $8 < -10 \tan^2 \theta + 30 \tan \theta - 10 < 12$.
- 1 mark:** Solves one complete case to find its boundary angles and interval solution.
- 1 mark:** Solves the second case and evaluating the correct overlapping compound range for θ .

Marker's comments:

- This question was poorly done by many candidates.
- Common mistake by candidates were treating each case strictly as a quadratic equation to find the roots.
While this identified the critical boundary angles, students heavily struggled to piece these values back into a logical final range, leading to a high failure rate for the final compound interval.
- Conversely, candidates who solved each case as a quadratic inequality were vastly more successful, as the inequality regions naturally guided them to the correct overlapping intersection.
- Candidates are reminded to leave their final answers in degrees to two decimal places as modelled in the solution, ensuring their responses align with physical trajectory contexts.

(c) (continued)

(iii) What is the total length of wet ground to the right of the board as θ varies?

2

Give your answer to the nearest metre.

To find the horizontal range x , let $y=0$

$$x \tan \theta - \frac{1}{90} x^2 \sec^2 \theta = 0$$

$$x \left(\tan \theta - \frac{1}{90} x \sec^2 \theta \right) = 0$$

Since the projectile of water lands at a distance $x \neq 0$:

$$\therefore \tan \theta - \frac{1}{90} x \sec^2 \theta = 0$$

$$x = \frac{90 \tan \theta}{\sec^2 \theta}$$

$$= 45 \times 2 \sin \theta \cos \theta$$

$$x = 45 \sin 2\theta$$

From c) ii), consider the 2 separate valid domains for θ

Case 1: $39.66^\circ < \theta < 51.93^\circ$

$$79.32^\circ < 2\theta < 103.86^\circ$$

As $-1 \leq \sin 2\theta \leq 1$ $\therefore 2\theta = 90^\circ$ gives $x_{\max}$ and (it lies within this interval)

$$\therefore x_{\max} = 45 \sin 90^\circ = 45 \text{ m}$$

The minimum distance for x occurs at the furthest bound from 90°

$$\therefore x_{\min} = 45 \sin(103.86^\circ) \doteq 43.69 \text{ m (2d.p.)}$$

Case 2: $59.88^\circ < \theta < 65.27^\circ$

$$\therefore 119.76^\circ < 2\theta < 130.54^\circ$$

Since $x = 45 \sin 2\theta$ is strictly decreasing in this domain

$$x_{\max} = 45 \sin(119.76^\circ) \doteq 39.06 \text{ m (2d.p.)}$$

$$x_{\min} = 45 \sin(130.54^\circ) \doteq 34.20 \text{ m (2d.p.)}$$

$$\therefore \text{Total length} = (45 - 43.69) + (39.06 - 34.20) \doteq 6 \text{ m (nearest metre)}$$

(c) (iii) (continued)

Marking Scheme:

1 mark: Derive the general horizontal range equation i.e. $x = 45 \sin 2\theta$ and correctly evaluates boundaries for Case 2.

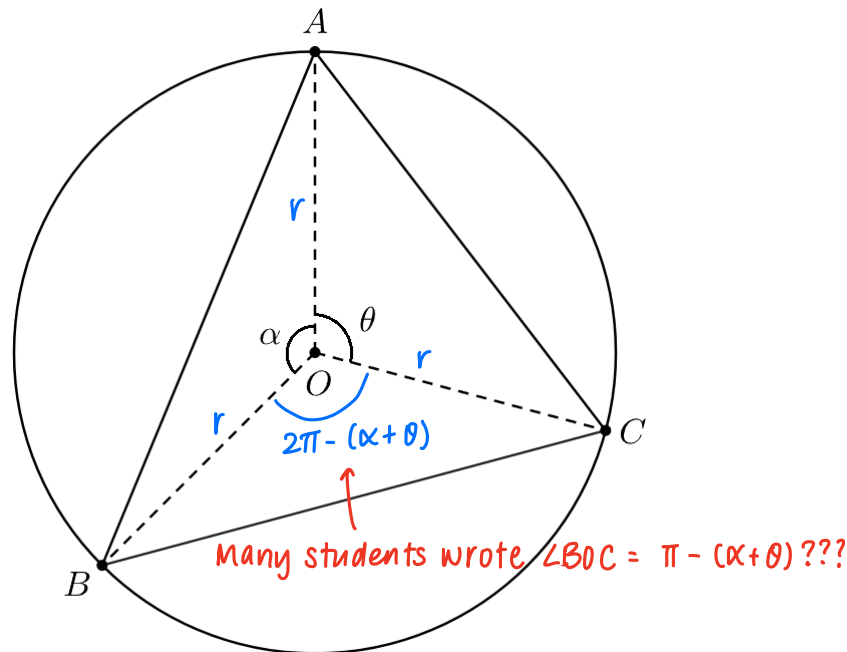
1 mark: Correctly accounts for the maximum at $2\theta = 90^\circ$ within Case 1, finds both interval lengths and obtains correct answer (i.e. $\approx 6m$).

Marker's comment:

- This question was exceptionally poorly done by candidates.
- Success in this question was dependent on the candidate's performance in part (ii), resulting in carried forward errors for most candidates.
- Out of the small group of candidates who was successful in part (ii), the majority still failed to get full marks as they didn't account for max distance occurring at $\theta = 45^\circ$, within Case 1, creating max distance of 45 m.

End of Q13 Solutions

(a) Let $\triangle ABC$ be inscribed in a circle of radius r centred at O .



Let $\angle AOB = \alpha$ and $\angle AOC = \theta$.

(i) Prove that the area of triangle ABC is given by

3

$$A = 2r^2 \sin\left(\frac{\alpha + \theta}{2}\right) \sin\frac{\alpha}{2} \sin\frac{\theta}{2}.$$

$\angle BOC = 2\pi - (\alpha + \theta)$ (angles at a point)

$$\begin{aligned} A &= \frac{1}{2}r^2 \sin\theta + \frac{1}{2}r^2 \sin\alpha + \frac{1}{2}r^2 \sin[2\pi - (\alpha + \theta)] \\ &= \frac{1}{2}r^2 [\sin\theta + \sin\alpha - \sin(\alpha + \theta)] \\ &= \frac{1}{2}r^2 \left[2 \sin\left(\frac{\theta + \alpha}{2}\right) \cos\left(\frac{\theta - \alpha}{2}\right) - 2 \sin\left(\frac{\theta + \alpha}{2}\right) \cos\left(\frac{\theta + \alpha}{2}\right) \right] \\ &= \frac{1}{2}r^2 \times 2 \sin\left(\frac{\theta + \alpha}{2}\right) \left[\cos\left(\frac{\theta - \alpha}{2}\right) - \cos\left(\frac{\theta + \alpha}{2}\right) \right] \\ &= \frac{1}{2}r^2 \times 2 \sin\left(\frac{\theta + \alpha}{2}\right) \times 2 \sin\frac{1}{2}\left(\frac{\theta + \alpha}{2} + \frac{\theta - \alpha}{2}\right) \sin\frac{1}{2}\left(\frac{\theta + \alpha}{2} - \frac{\theta - \alpha}{2}\right) \\ &= \frac{1}{2}r^2 \times 2 \sin\left(\frac{\theta + \alpha}{2}\right) \times 2 \sin\frac{\theta}{2} \sin\frac{\alpha}{2} \\ &= 2r^2 \sin\left(\frac{\theta + \alpha}{2}\right) \sin\frac{\alpha}{2} \sin\frac{\theta}{2} \end{aligned}$$

- Prove question = Show ALL working. Students cannot expect markers to read their mind.
- Students need to revise their trigonometric identities.

(a) (continued)

- (ii) Hence, or otherwise, find the value of θ that maximises the area of the triangle if α is fixed.

2

Method 1

$$A = 2r^2 \sin \frac{\alpha}{2} \left[\sin \left(\frac{\theta + \alpha}{2} \right) \sin \frac{\theta}{2} \right]$$

$$= 2r^2 \sin \frac{\alpha}{2} \times \frac{1}{2} \left[\cos \frac{\alpha}{2} - \cos \left(\theta + \frac{\alpha}{2} \right) \right] \quad \text{Note: } \sin A \sin B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$= r^2 \sin \frac{\alpha}{2} \left[\cos \frac{\alpha}{2} - \cos \left(\theta + \frac{\alpha}{2} \right) \right]$$

$\therefore \alpha$ is constant, maximum occurs when $\cos \left(\theta + \frac{\alpha}{2} \right) = -1$.

$$\theta + \frac{\alpha}{2} = \pi$$

$$\theta = \pi - \frac{\alpha}{2}$$

Method 2

$$A = 2r^2 \sin \frac{\alpha}{2} \left[\sin \left(\frac{\theta + \alpha}{2} \right) \sin \frac{\theta}{2} \right]$$

$$\frac{dA}{d\theta} = 2r^2 \sin \frac{\alpha}{2} \left[\frac{1}{2} \cos \left(\frac{\theta + \alpha}{2} \right) \sin \frac{\theta}{2} + \frac{1}{2} \cos \frac{\theta}{2} \sin \left(\frac{\theta + \alpha}{2} \right) \right]$$

$$= r^2 \sin \frac{\alpha}{2} \sin \left(\frac{\theta + \alpha}{2} + \frac{\theta}{2} \right)$$

$$= r^2 \sin \frac{\alpha}{2} \sin \left(\theta + \frac{\alpha}{2} \right) = 0$$

$$\sin \left(\theta + \frac{\alpha}{2} \right) = 0$$

$$\theta + \frac{\alpha}{2} = 0, \pi, 2\pi, \dots$$

$$\theta = -\frac{\alpha}{2}, \pi - \frac{\alpha}{2}, 2\pi - \frac{\alpha}{2}, \dots$$

$$\therefore \theta \in [0, \pi]$$

$$\therefore \theta = \pi - \frac{\alpha}{2}$$

$$\frac{d^2A}{d\theta^2} = r^2 \sin \frac{\alpha}{2} \cos \left(\theta + \frac{\alpha}{2} \right)$$

$$\frac{d^2A}{d\theta^2} \left(\pi - \frac{\alpha}{2} \right) = r^2 \sin \frac{\alpha}{2} \cos \pi$$

$$= -r^2 \sin \frac{\alpha}{2} < 0 \quad \therefore \text{Maximum at } \theta = \pi - \frac{\alpha}{2}.$$

- Poorly done.
- Some students found θ , but didn't prove it was a maximum.
- Method 1 was more efficient, but unfortunately not very popular.

(a) (continued)

- (iii) Hence, by treating θ as a fixed angle or otherwise, find the values of α and θ where the maximum area occurs and state the type of triangle formed. 2

Similarly, $\alpha = \pi - \frac{\theta}{2}$

$$\theta = \pi - \frac{\alpha}{2} \quad (\text{from ii})$$

$$= \pi - \frac{1}{2} \left(\pi - \frac{\theta}{2} \right)$$

$$= \pi - \frac{\pi}{2} + \frac{\theta}{4}$$

$$\frac{3\theta}{4} = \frac{\pi}{2}$$

$$\theta = \frac{2\pi}{3}$$

$$\alpha = \pi - \frac{1}{2} \left(\frac{2\pi}{3} \right) = \frac{2\pi}{3}$$

$$\angle BOC = \frac{2\pi}{3} \quad (\text{angles at a point})$$

$$\begin{aligned} \angle ABO = \angle OBC &= \frac{1}{2} \left(\pi - \frac{2\pi}{3} \right) \quad (\text{angle sum of triangle}) \\ &= \frac{\pi}{6} \end{aligned}$$

$$\therefore \angle ABC = \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3} \quad (\text{angle sum})$$

$$\text{Similarly, } \angle BAC = \angle ACB = \frac{\pi}{3}$$

OR $\angle ABC = \frac{\pi}{3}$ (angle at centre and circumference)

$$\therefore \triangle ABC \text{ is equilateral (all angles } \frac{\pi}{3})$$

- Very poorly done.
- Some students were lucky enough to guess $\triangle ABC$ is equilateral.
- Many students wrote $\theta = \frac{\pi}{3} = \alpha$, this would mean $\angle BAC = \frac{2\pi}{3}$, so angle sum of triangle would be $\frac{2\pi}{3} \times 3 = 2\pi$.

- (b) There is a chemistry course at the University of Old North Scotland that has a course size of 300 students. Students cancel their enrolment right before the course starts which leads to empty spots. Therefore, the lecturer allows for more than 300 enrolments before the course to account for this, given that the lecturer expects to be overbooked at most 1% of the time. What is the range of values of the probability that a singular student comes to class if the lecturer opens up 320 enrolments?

4

You may assume that the sample size is sufficient for the normal approximation to the binomial distribution to be used (without continuity correction).

You are given that if $Z \sim N(0, 1)$ then $P(Z < 1.285) = 0.9$ and $P(Z < 2.326) = 0.99$.

$$P(X > 300) < 0.01$$

$$\mu = 320p$$

$$P(X \leq 300) > 0.99$$

$$\sigma = \sqrt{320p(1-p)}$$

$$P\left(Z \leq \frac{300 - 320p}{\sqrt{320p(1-p)}}\right) > 0.99$$

$$\frac{300 - 320p}{\sqrt{320p(1-p)}} > 2.326 \rightarrow 300 - 320p > 0$$

$$300 > 320p$$

$$p < \frac{300}{320} = 0.9375$$

$$300 - 320p > 2.326 \sqrt{320p(1-p)}$$

$$(300 - 320p)^2 > [2.326 \sqrt{320p(1-p)}]^2$$

$$90000 - 192000p + 102400p^2 > 2.326^2 \times 320p(1-p)$$

$$\text{Let } a = 2.326^2 \times 320$$

$$p^2(102400 + a) - p(192000 + a) + 90000 > 0$$

$$p = \frac{192000 + a \pm \sqrt{(192000 + a)^2 - 4(102400 + a)(90000)}}{2(102400 + a)}$$

$$p = 0.9622743618, 0.8981778 \dots$$

$$\therefore p < 0.9375$$

$$\therefore 0 < p < 0.8982 \text{ (4dp)}$$

(b) (continued)

- Terribly done...
- Students wrote $n = 300$.
- Many didn't know where to start.

- $z = \frac{x - \mu}{\sigma}$

$$\sigma = \sqrt{\text{Var}(x)}$$

Some students wrote: $\frac{300 - 320p}{320p(1-p)} > 2.326$. This made the question significantly easier.

- Expect wordy questions for this topic. See 2022 ME1 Q14d for similar Q. Students need to be able to breakdown the question and extract key information.

(c) (i) Prove that if $\dot{w} = lw$, then $w = Ae^{lt}$ for some real numbers A and l .

1

$$\frac{dw}{dt} = lw$$

$$\int \frac{dw}{w} = \int l dt$$

$$\ln|w| = lt + C$$

$$|w| = e^{lt+C}$$

$$w = \pm e^{lt} e^C$$

* must have explicit mention of $\pm$ *

$$\therefore w = Ae^{lt} \text{ where } A = \pm e^C$$

• Poorly done, many didn't read the question carefully.

• If $\dot{w} = lw$, then $w = Ae^{lt}$.
start with this/ prove this
can assume this

Many students answered: If $w = Ae^{lt}$, then $\dot{w} = lw$.

(c) (continued)

(ii) Suppose that y_1 and y_2 are the solutions to the paired differential equations

1

$$\dot{y}_1 = 2y_1 + 6y_2,$$

$$\dot{y}_2 = 2y_1 + 3y_2$$

with the initial conditions $y_1(0) = 2$ and $y_2(0) = -3$.

Prove that $\dot{y}_1 + k\dot{y}_2 = (2 + 2k)y_1 + (6 + 3k)y_2$ for all real constants k .

$$\text{LHS} = 2y_1 + 6y_2 + k(2y_1 + 3y_2)$$

$$= y_1(2 + 2k) + y_2(6 + 3k)$$

$$= \text{RHS}$$

- This question should be an easy one mark for all, yet some still couldn't get it.

(c) (continued)

- (iii) Hence, by letting $k = 2$ and another suitable value, solve the differential equations for y_1 and y_2 . 4

$$\text{If } \dot{w} = \ell w, \text{ then } w = Ae^{\ell t}$$

$$\dot{w} = \dot{y}_1 + k\dot{y}_2 = \ell(y_1 + ky_2) = (2+2k)y_1 + (6+3k)y_2$$

$$\begin{aligned} \text{Let } k=2, \quad \dot{y}_1 + 2\dot{y}_2 &= (2+2 \times 2k)y_1 + (6+3 \times 2)y_2 \\ &= 6y_1 + 12y_2 \\ &= 6(y_1 + 2y_2) \end{aligned}$$

$$\therefore \ell = 6$$

$$\therefore w = y_1 + 2y_2 = Ae^{6t}$$

$$\begin{aligned} t=0, y_1=2, y_2=-3: 2-6 &= A \\ A &= -4 \end{aligned}$$

$$\therefore \ell(y_1 + ky_2) = (2+2k)y_1 + (6+3k)y_2$$

Equating coefficients of y_1 and y_2

$$\ell = 2+2k \quad \text{①}$$

$$\ell k = 6+3k \quad \text{②}$$

$$\text{①} \Rightarrow \text{②}: (2+2k)k = 6+3k$$

$$2k^2 + 2k - 3k - 6 = 0$$

$$2k^2 - k - 6 = 0$$

$$(2k+3)(k-2) = 0$$

$$k = -\frac{3}{2}, 2$$

$$\begin{aligned} \text{Let } k = -\frac{3}{2}, \quad \dot{y}_1 - \frac{3}{2}\dot{y}_2 &= (2+2 \times -\frac{3}{2})y_1 + (6+3 \times -\frac{3}{2})y_2 \\ &= -y_1 + \frac{3}{2}y_2 \\ &= -(y_1 - \frac{3}{2}y_2) \end{aligned}$$

$$\therefore \ell = -1$$

$$\therefore w = y_1 - \frac{3}{2}y_2 = Be^{-t}$$

$$\begin{aligned} t=0, y_1=2, y_2=-3: 2 - \frac{3}{2}(-3) &= B \\ B &= \frac{13}{2} \end{aligned}$$

(c) (continued)

$$\therefore y_1 + 2y_2 = -4e^{6t} \quad (3)$$

$$y_1 - \frac{3}{2}y_2 = \frac{13}{2}e^{-t} \quad (4)$$

$$(3) - (4) : \frac{7}{2}y_2 = -4e^{6t} - \frac{13}{2}e^{-t}$$

$$y_2 = -\frac{2}{7}(4e^{6t} + \frac{13}{2}e^{-t})$$

$$= -\frac{8}{7}e^{6t} - \frac{13}{7}e^{-t} \rightarrow (3)$$

$$y_1 + 2(-\frac{8}{7}e^{6t} - \frac{13}{7}e^{-t}) = -4e^{6t}$$

$$y_1 = -4e^{6t} + \frac{16}{7}e^{6t} + \frac{26}{7}e^{-t}$$

$$= -\frac{12}{7}e^{6t} + \frac{26}{7}e^{-t}$$

- Horribly done.
- Either due to lack of time or understanding, not many marks were awarded.
- Students needed to use (i) and (ii) to help.

End of Q14 Solutions

Section I

Multiple Choice Solutions

1	2	3	4	5	6	7	8	9	10
A	D	D	A	B	C	B	D	C	A

1 Which of the following is equal to $\int \frac{1}{\sqrt{4-9x^2}} dx$?

A. $\frac{1}{3} \sin^{-1} \frac{3x}{2} + C$

B. $\frac{1}{6} \sin^{-1} \frac{3x}{2} + C$

C. $\frac{1}{3} \sin^{-1} \frac{x}{2} + C$

D. $\frac{1}{6} \sin^{-1} \frac{x}{2} + C$

Method 1:
$$\int \frac{1}{\sqrt{4-9x^2}} dx = \int \frac{1}{3\sqrt{\frac{4}{9}-x^2}} dx$$

$$= \frac{1}{3} \int \frac{1}{\sqrt{\frac{4}{9}-x^2}} dx$$

$$= \frac{1}{3} \sin^{-1} \left(\frac{3x}{2} \right) + C$$

Method 2: Reference sheet

$$\int \frac{f'(x)}{\sqrt{1-[f(x)]^2}} = \sin^{-1} f(x) + C$$

$$\int \frac{1}{\sqrt{4-9x^2}} dx = \frac{1}{3} \int \frac{3}{\sqrt{4-(3x)^2}} dx$$

$$= \frac{1}{3} \sin^{-1} \left(\frac{3x}{2} \right) + C$$

A	146
B	2
C	5
D	0

2

A 100 litre tub of salt water has initial concentration 3 g/L.

Salt water of concentration 2 g/L flows in at 1 L/min and the mixture is stirred continuously, before flowing out of the tub at 1 L/min. Which of the following correctly gives the equation for the rate of change of concentration c in g/L at time t minutes?

A. $\frac{dc}{dt} = 2 - \frac{c}{100}$

B. $\frac{dc}{dt} = \frac{1}{100} - 2c$

C. $\frac{dc}{dt} = \frac{1}{50} - c$

D. $\frac{dc}{dt} = \frac{1}{50} - \frac{c}{100}$

If $c = 2$ then the system is in equilibrium i.e. same in as same out.
This is only true for option D.

Otherwise:

Input:

$$\text{Grams in} = 2 \text{ g/L} \times 1 \text{ L/min} = 2 \text{ g/min}$$

$$\therefore \frac{dc}{dt} = \frac{2}{100}$$

Output:

$$\text{Grams out} = c \text{ g/L} \times 1 \text{ L/min} = c \text{ g/min}$$

$$\therefore \frac{dc}{dt} = \frac{c}{100}$$

Input – Output:

$$\frac{dc}{dt} = \frac{2}{100} - \frac{c}{100} = \frac{1}{50} - \frac{c}{100}$$

A	66
B	10
C	27
D	50

- 3 Let $P(x)$ be a polynomial that leaves a remainder of 2 when divided by $x - 3$ and a remainder of -2 when divided by $x + 1$.

Which of the following is the remainder of $P(x)$ when it is divided by $x^2 - 2x - 3$?

A. $-x + 1$

B. $-x - 1$

C. $x + 1$

☒ D. $x - 1$

By the Remainder Thm:

$$P(3) = 2$$

$$P(-1) = -2$$

Note: $x^2 - 2x - 3 = (x - 3)(x + 1)$

$$P(x) = (x - 3)(x + 1)Q(x) + mx + c$$

$$[\because \deg(\text{remainder}) < \deg(\text{divisor})]$$

$$\therefore 3m + c = 2$$

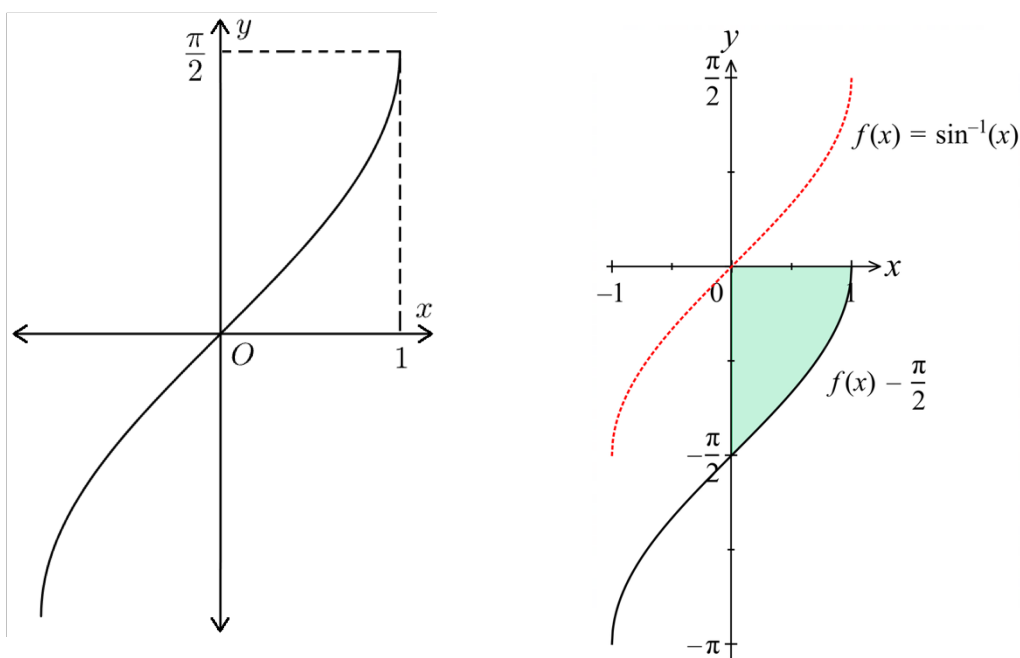
$$-m + c = -2$$

$$\therefore 4m = 4 \Rightarrow m = 1$$

$$\therefore c = -1$$

A	11
B	12
C	9
D	120

- 4 The graph shows a sketch of $y = \sin^{-1} x$.



The graph is shifted down by $\frac{\pi}{2}$ units, before rotating around the region bounded by the curve and $y = 0$, around the x -axis to form a volume of revolution.

Which of the following is equal to the volume of the solid?

☒ A. $\pi \int_0^1 \left(\frac{\pi}{2} - \sin^{-1} x \right)^2 dx$

B. $\pi \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \sin x \right)^2 dx$

C. $\pi \int_0^1 \left[\frac{\pi}{2} - (\sin^{-1} x)^2 \right] dx$

D. $\pi \int_0^{\frac{\pi}{2}} \left[\frac{\pi}{2} - (\sin x)^2 \right] dx$

$$\text{Volume} = \pi \int_0^1 f(x)^2 dx$$

$$\text{The 'new' function is } f(x) = \sin^{-1}(x) - \frac{\pi}{2}$$

Or, students can note after drawing the transformation that the other responses are clearly invalid/wrong.

A	120
B	12
C	15
D	6

- 5 A test contains 5 questions, each question can score 1 point if it is answered correctly, 0 if not answered, and will deduct a point from the total score if it is answered incorrectly.

The probability of a student answering a random question correctly is 0.3.

The student will also leave on average, one out of ten questions blank.

A score of at least 3 is required to pass.

What is the probability that the student will pass?

A. 0.00837

☒ B. 0.03348

C. 0.06615

D. 0.1638

$P(\text{correct}) = 0.3$; $P(\text{left blank}) = 0.1$; $P(\text{incorrect}) = 0.6$

Score = 5: 5 correct answers
 $P(5 \text{ correct}) = 0.3^5$

Score = 4: 4 correct + 1 blank or 4 correct + 1 wrong
 5C_1 to get the 4 correct responses
 $P(4 \text{ correct}) = {}^5C_1 \times 0.3^4 \times (0.6 + 0.1) = {}^5C_1 \times 0.3^4 \times 0.7$

Score = 3: 3 correct + 2 blanks
 5C_2 to get the 3 correct responses
 $P(3 \text{ correct}) = {}^5C_2 \times 0.3^3 \times 0.1^2$

A	20
B	80
C	39
D	14

- 6 Which of the following is equivalent to the function $f(\theta) = \cos^{-1}(\cos \theta)$ defined over the domain $\theta \in [0, 2\pi]$?

A. $\pi + |\pi - \theta|$

B. $\pi + |\pi + \theta|$

☒ C. $\pi - |\pi - \theta|$

D. $\pi - |\pi + \theta|$

A	14
B	3
C	124
D	12

For $\theta \in [0, \pi]$: $f(\theta) = \theta$

Only option C gives this.

Note: $|\pi - \theta| = \pi - \theta$ as $0 \leq \theta \leq \pi$

- 7 9 position vectors all have length 6 units. At least two of the position vectors will have a minimum dot product equal to which of the following?

A. 0

☒ B. $18\sqrt{2}$

C. $18\sqrt{3}$

D. 36

A	28
B	81
C	38
D	6

These 9 vectors lie on a circle, radius 6 units, centred at the origin.

Divide the circle into 8 equal sectors, with angular separation of 45° .

By the PHP, 2 of the vectors must lie in the same sector (or the boundary of two sectors)

Note: $\cos \theta$ is decreasing for $0 \leq \theta \leq \frac{\pi}{4}$

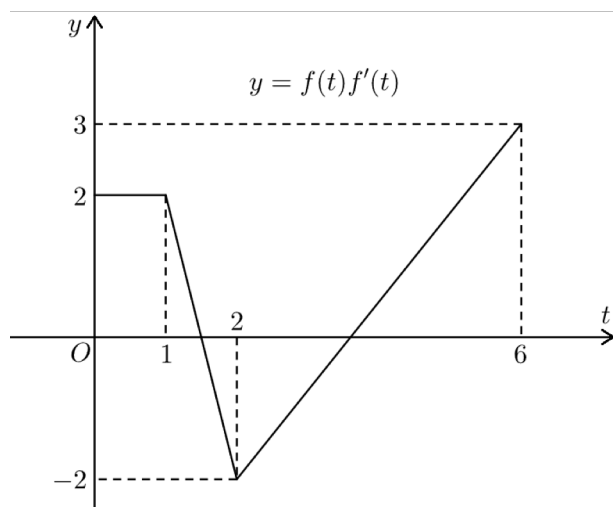
$$\mathbf{v}_i \cdot \mathbf{v}_j = |\mathbf{v}_i| \times |\mathbf{v}_j| \cos \theta$$

$$\geq 6 \times 6 \times \frac{1}{\sqrt{2}}$$

$$= 18\sqrt{2}$$

8 A particle's velocity is a function of time $v = f(t)$.

The diagram shows the graph of $y = f(t)f'(t)$.



The particle has an initial velocity of 3 m/s.

Calculate the speed of the particle at time $t = 6$.

A. 3 m/s

B. 4 m/s

C. $\sqrt{13}$ m/s

D. $\sqrt{17}$ m/s

A	11
B	42
C	56
D	44

Note by the reverse chain rule:
$$\int_a^b f(t) f'(t) dt = \left[\frac{1}{2} f(t)^2 \right]_a^b$$

What is the equation of the line for $2 < t < 6$?
$$y - 3 = \frac{3 - (-2)}{6 - 2} (t - 6)$$

$$\therefore y = 3 + \frac{5}{4}(t - 6)$$

$$\int_2^6 f(t) f'(t) dt = \left[3t + \frac{5}{8} (t-6)^2 \right]_2^6 = 18 - (6 + 10) = 2$$

$$\int_0^2 f(t) f'(t) dt = 2 + 0 = 2 \Rightarrow \int_0^6 f(t) f'(t) dt = 2 + 2 = 4$$

$$\therefore \frac{1}{2} [v(6)^2 - v(0)^2] = 4.5 \Rightarrow v(6)^2 - 9 = 8$$

$$\therefore |v(6)| = \sqrt{17}$$

9 Let $f(x) = (2x + 1)(3x + 1)(4x + 1) \dots (20x + 1)$.

Which of the following is equal to $f'(0)$?

A. 207

B. 208

C. 209

D. 210

A	3
B	17
C	117
D	16

Method 1: Look for a pattern i.e. inductive thinking (and the product rule)

$$n = 1: f_1'(0) = 2$$

$$n = 2: f_2'(0) = 2 \times 1 + 3 \times 1 = 5 = 2 + 3$$

$$n = 3: f_3'(0) = 5 \times 1 + 4 \times 1 = 9 = 5 + 4 = 2 + 3 + 4$$

$$n = 4: f_4'(0) = 9 \times 1 + 5 \times 1 = 14 = 9 + 5 = 2 + 3 + 4 + 5$$

$$\text{So } f'(0) = 2 + 3 + 4 + \dots + 20 = \frac{22 \times 19}{2} = 209$$

Method 2: Implicit differentiation

Take $\ln$ ($\log_e$) of both sides and let $y = f(x)$

$$\ln y = \ln (2x + 1)(3x + 1)(4x + 1) \times \dots \times (20x + 1)$$

$$= \ln (2x + 1) + \ln (3x + 1) + \ln (4x + 1) + \dots + \ln (20x + 1)$$

Differentiation both sides wrt x :

$$\frac{y'}{y} = \frac{2}{2x + 1} + \frac{3}{3x + 1} + \frac{4}{4x + 1} + \dots + \frac{20}{20x + 1}$$

$$\therefore \frac{y'(0)}{y(0)} = 2 + 3 + 4 + \dots + 20 \Rightarrow y'(0) = (2 + 3 + 4 + \dots + 20) y(0)$$

$$\therefore y(0) = 1 \Rightarrow y'(0) = 2 + 3 + 4 + \dots + 20 = \frac{22 \times 19}{2} = 209$$

10 Let $y = f(x)$ be a continuous function defined for all real x .

It has an inverse function defined by $y = g(x)$.

Which of the following is equal to the derivative of $y = g(g(x))$?

A. $\frac{1}{f'(g(x))f'(g^2(x))}$

B. $\frac{1}{f'(x)f'(g(x))}$

C. $\frac{1}{g'(f(x))g'(f^2(x))}$

D. $\frac{1}{g'(x)g'(f(x))}$

A 48

B 52

C 23

D 29

$$g(x) = f^{-1}(x)$$

$$\therefore g'(x) = (f^{-1})'(x)$$

$$= \frac{1}{f'(f^{-1}(x))}$$

$$= \frac{1}{f'(g(x))}$$

In general: $\frac{d}{dx} g(g(x)) = g'(x) \times g'(g(x))$

$$\therefore \frac{d}{dx} g(g(x)) = \frac{1}{f'(g(x))} \times \frac{1}{f'(g(g(x)))}$$

$$= \frac{1}{f'(g(x))} \times \frac{1}{f'(g^2(x))}$$

$$= \frac{1}{f'(g(x))f'(g^2(x))}$$